# When Do Taxpayers Want a Tax Inspector?

Personal Interaction, Private Concessions and Correction

Ahsan Zia Farooqui

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**Abstract**

A fundamental challenge in public finance is how governments can finance public services by raising revenue through tax fairly and effectively. Revenue authorities pursue these State objectives by implementing tax rules through frontline officers who assess liabilities, explain obligations, and help resolve disputes. Frontline tax officers can enforce tax rules on behalf of the State, but they may also offer preferential treatment, making personal interaction a source of both assistance and private advantage. These interactions may lead officers to use considerable discretion for their private rents, creating demand for technology-based, impersonal tax administration. While technology may replace some functions of tax officers, it cannot eliminate the need for human judgment entirely. What do citizens value in personal tax administration? I answer this question by running a conjoint experiment with taxpayers in Lahore, Pakistan. I find that 60% of the respondents prefer technology-based assessment of their tax liability. However, choosing the satellite does not imply rejecting private deals with tax officers. Between two inspectors, 58.5% of respondents who initially preferred the satellite to assess their tax liability preferred a scenario where the inspector was hypothetically shown offering to lower the tax bill in exchange for an informal payment. This association persists after accounting for trust in officers' fairness. These findings suggest that citizens' demand for personal tax administration reflects distinct expectations about assessment, assistance, and private advantage. Understanding these expectations matters for building trust in revenue authorities, which can support voluntary compliance and fair collection of revenue to finance public services.

**JEL:** H26, H71, D73, O33

# 1 Introduction

A central problem in public finance is how governments can raise funds for public services and redistribute resources while limiting their economic and administrative costs. The capacity of any State to raise revenue depends on the institutions that translate statutory rules into effective taxation, as well as on economic development and political choices (Besley and Persson, 2014). While research identifies the importance of deterrence measures as a tool to enforce tax compliance (Slemrod, 2019), beyond these measures, taxpayers' perception of tax authority itself is an important driver of compliance (Luttmer and Singhal, 2014). Murphy (2004) explores how trust plays an important role in the relationship between the tax authorities and taxpayers accused of tax avoidance, while van Dijke and Verboon (2010) find that procedural fairness is particularly consequential for voluntary compliance among citizens with low trust in authorities. Similarly, Kirchler et al. (2008) provide a formal framework which distinguishes between trust-based cooperation and enforced compliance. These contributions establish the relevance of the taxpayer–authority relationship to tax collection. For administrative reform, however, an overall judgment of trust leaves an important question unresolved: which aspects of that relationship do taxpayers want to retain, and which would they prefer to replace?

This question arises because personal tax administration bundles functions that taxpayers may evaluate differently. An inspector can establish taxable facts, explain a liability, and help correct a record. The same discretion can also enable preferential treatment in exchange for an informal payment. A taxpayer who distrusts an inspector to assess a property fairly may nevertheless value access to that inspector when a bill is disputed. Conversely, willingness to accept an inspector's financial offer need not imply confidence in impartial administration. The distinction has direct fiscal significance. Correcting an excessive assessment improves the application of the intended tax rules, whereas negotiating a payment below the correct liability benefits the taxpayer at the expense of public revenue. Both can make personal interaction attractive, but they imply different reasons for preserving or restricting official discretion.

These distinctions become particularly consequential as revenue authorities introduce technologies that change the role of frontline officials. Technology is changing the function of public agencies from "street-level" to "systems-level" bureaucracy where algorithms make decisions about clients based on data collected (Bovens and Zouridis, 2002). This means that tasks performed by frontline officers' roles are also being redefined in the "systems-level" bureaucratic regimes and the current research identifies potential substitution or complementarity of frontline officers and technology (Autor et al., 2003; Bullock, 2019). Current research identifies less complex routine tasks that apply explicit rules, such as applying statutory rules to determine the tax base, as more likely to be replaced by technology (Bullock, 2019). However, complex tasks involving communication and problem-solving may still need officers' judgment and local knowledge (Gräfe et al., 2024). In taxation, this may mean that reducing personal contact through technology can limit officer rents, but it can also change access to more complex officer tasks that citizens previously used to navigate the system. Understanding demand for im-

personal technology-based tax administration therefore requires distinguishing preferences over the assessor from preferences over assessment outcomes, financial transactions, and access to recourse.

I study these preferences for urban property-tax administration in Lahore, the capital of Punjab, Pakistan. Inspectors play a central role in establishing the property characteristics used to calculate liabilities, communicating with taxpayers, and initiating amendments to the administrative record. Their recommendations require supervisory approval, but their responsibility for producing property-specific information makes them central to taxpayers' encounters with the authority. Lahore also provides a relevant reform environment: Punjab has introduced GIS-based administrative changes and shifted property valuation from annual rental value towards capital value based on administrative rates. Among taxpayers in my survey who provide a substantive response, approximately half report little or no trust in officers to value properties fairly. This dissatisfaction makes alternatives to inspector assessment salient. However, it remains unclear whether taxpayers favour replacing tax officers entirely with technology-based administration and whether they expect errors under such systems to be readily corrected.

What determines taxpayers' willingness to choose impersonal tax assessment, beyond the amount they ultimately pay? I investigate this question through three related comparisons. First, do taxpayers prefer an inspector or a satellite-based system when both produce the correct bill at the same cost? Second, how do initial assessment errors and the terms of paid concessions shape choices over assessment arrangements? Third, how do these choices relate to trust in officers' valuation fairness and expectations about correcting computer-generated errors? These comparisons connect the design of tax administration to the particular outcomes and institutional expectations underlying taxpayers' choices.

I combine a survey experiment with 2,100 property taxpayers, linked administrative records, and independent remote-sensing measurements. In the experiment, taxpayers repeatedly choose between assessment receipts displaying the assessor, a reference correct liability, the initial bill, any informal payment, the final official bill, and total expenditure. Inspector offers can include a paid concession; satellite offers do not. Four common comparisons precede Bayesian adaptive choice tasks, which use earlier responses to select subsequent comparisons for their expected information gain. The analysis retains 21,001 paired choices and estimates preferences conditional on the sequential design. The linked administrative records document assessment and correction activity, while survey measures capture officer trust, expected correctability, and obstacles to seeking correction. A separate offer of computer-estimated information about respondents' own properties provides evidence on demand for independent property information.

A simple choice framework separates the valuation of the assessor, the initial assessment state, and entering a paid transaction from the cash cost of the selected arrangement. This separation matters because the same final expenditure can arise from different administrative processes. A correct initial assessment and an excessive assessment followed by a reduction need not be equally attractive, even when they leave the taxpayer paying the same amount. Similarly, a paid concession may carry additional costs or benefits beyond its net financial saving. The

framework makes it possible to examine whether financial savings alone account for choices, whether transaction valuations depend on the initial assessment, and whether errors change the relative attractiveness of an inspector. I estimate a pooled choice model, a panel mixed model allowing persistent preference heterogeneity, and a specification restricted to the later adaptive tasks.

Three sets of findings emerge. First, taxpayers exhibit substantial variation in assessor choice even when the stated financial outcomes are identical. In the opening comparison, 59.8% choose satellite assessment and 40.2% choose the inspector. This is a majority for the satellite offer as presented, although its position is fixed and the position-adjusted average assessor coefficient is imprecisely estimated. A preference for the satellite in this comparison also does not imply rejection of private transactions: among eligible initial satellite choosers, 58.5% subsequently accept a paid inspector concession over an unpaid inspector offer. The latter comparison is between two inspector offers, rather than a direct choice between a concession and satellite assessment. Together, the comparisons distinguish assessor preference on equal terms from willingness to accept a financially advantageous transaction when the alternatives change.

Second, choices reflect more than eventual expenditure. Initially incorrect assessments are generally less attractive after accounting for final expenditure, and paid-transaction terms contribute to choice beyond their financial consequences. The restriction that paid concessions matter only through total expenditure is rejected across the main specifications. However, evidence that the initial assessment changes transaction valuation depends on the specification, and the results do not establish that overassessment increases the relative attractiveness of inspectors. The more consistent finding is that taxpayers distinguish assessment and settlement arrangements that cannot be ranked by final cost alone. These estimates describe net valuations of the stated arrangements; they do not assign those valuations a unique psychological explanation.

Third, expectations about recourse are strongly associated with demand for impersonal assessment. At equal correct bills, satellite choice rises from 41.6% among taxpayers who expect correcting a computer error to be very hard, to 62.0% among those expecting it to be hard, and 89.6% among those expecting it to be easy or very easy. The association persists across the repeated choices after accounting for officer trust and allowing choice coefficients to differ across recorded information groups. Expected correctability is not randomly assigned, so these comparisons do not identify the effect of providing a correction channel. They nevertheless show that demand for satellite assessment differs substantially with beliefs about recourse, even when the displayed assessment is correct. Complementary evidence indicates demand for independent property information: approximately 39% accept the information offer at each of the two positive prices.

The paper contributes to research on taxpayer cooperation by connecting specific institutional beliefs to choices over tax administration. The literature on tax morale and trust establishes the importance of taxpayers' relationship with the state (Luttmer and Singhal, 2014; Kirchler et al., 2008), while subsequent work distinguishes forms of trust and examines how interactions with

officials shape taxpayer perceptions (Gangl et al., 2015; Mascagni et al., 2025). I examine a related but distinct question: how taxpayers choose between assessment arrangements when the assessor, initial assessment, and financial terms are made explicit. This approach distinguishes confidence in officers' fairness from beliefs about correcting computer errors, and relates both to assessor choice. It provides evidence on why a general expression of dissatisfaction is insufficient to predict the administrative arrangements a taxpayer will select.

The paper also contributes to public-finance research on discretion, collector incentives, and digital tax administration. Khan et al. (2016) examine how incentives facing collectors change revenue collection and informal payments. Okunogbe and Pouliquen (2022) study how replacing in-person filing changes compliance costs and tax payments, while Knebelmann et al. (2026) examine the accuracy, equity, and acceptability of algorithmic property valuation. I complement this work by separating taxpayer responses to assessor identity, initial assessment errors, and paid concessions within a common choice framework, and by examining their relationship with expected recourse. The distinction between private attractiveness and fiscal desirability is central: an arrangement can appeal to a taxpayer because it supports accurate taxation or because it permits a favourable departure from the rules.

For revenue authorities, the findings make the arrangements surrounding assessment a salient part of technological reform. Taxpayers may evaluate an impersonal system partly through their expectations about what happens when its decisions are wrong. At the same time, acceptance of a private concession cannot be treated as evidence that discretionary bargaining should be preserved. The relevant design question is how to provide dependable assessment and accessible correction while limiting preferential treatment. This paper supplies evidence on taxpayers' choices and the beliefs associated with them; establishing how alternative arrangements affect compliance, revenue, and welfare requires evidence from their implementation.

The remainder of the paper is organised as follows. Section 2 describes the institutional setting and the taxpayer–authority relationship. Section 3 develops the choice framework. Sections 4–6 present the data, outcome measures, and empirical strategy. Section 7 reports the results on assessor preferences, assessment errors, paid concessions, demand for independent evidence, and expected correctability. Section 8 concludes.

# 2 Context

## 2.1 Study Site

The study site is Lahore, the provincial capital of the province of Punjab, Pakistan. With a population of 13 million people, Lahore is the world's 15th largest city. For getting access to the administrative data, I partnered with Punjab's Excise and Taxation (E&T) Department, a provincial government revenue authority that administers the collection and billing of the

primary property tax in metropolitan cities. In Lahore, E&T administers the taxation of almost 1 million properties in Lahore.

## 2.2 Property-value Assessment and the Role of Front-line Tax Officers

Punjab's Urban Immovable Property Tax (UIPT) is administered by the provincial Excise and Taxation Department. The department divides the urban jurisdictions subject to UIPT into a five-level geographical hierarchy, illustrated in Figure A.1. The primary unit for property inspection, assessment, collection, and enforcement is called the tax "circle". Each circle is assigned an inspector, supported by a clerk and a constable. The inspector leads property visits, determines the relevant property characteristics, and takes primary responsibility for the recommended assessment. The clerk and constable assist with measurement, documentation, record preparation, and the delivery of notices. Accordingly, the term "officer" in the theoretical analysis refers to the inspector as the relevant front-line decision-maker.

The circle team cannot enter its recommendation directly into the official tax record. It must submit the assessment to the Excise and Taxation Officer (ETO), whose approval is mandatory. The ETO reviews the measurements, classifications, and supporting record prepared by the inspector-led team. The process therefore separates informational control from formal authority: the inspector controls the production of property-specific information, while the ETO controls whether the recommendation receives formal approval. For analytical tractability, the model assumes that the ETO is not a party to any informal bargain between the inspector and the taxpayer. ETO approval nevertheless matters because the inspector must produce a record that is sufficiently complete and internally consistent to obtain formal approval.

## 2.3 The Taxpayer–Administration Relationship

A recent survey reveals the taxpayer– administration relationship in the study site. It distinguishes three related but conceptually separate dimensions of the fiscal relationship: perceived diversion during collection, trust in the institutions administering the tax and perceptions of the return from taxation. These distinctions matter because dissatisfaction with one dimension does not reveal which features of personal administration taxpayers would prefer to retain or remove.

Table 1 reports these measures. On average, citizens attribute 62.2% of the gap to informal payments, and 71.0% assign more than half of the gap to this explanation. These responses indicate a widespread perception that private transactions divert revenue from the formal collection process.

**Table 1:** Baseline surveys: perceptions of collection, trust and public services

| Measure | Count / respondents | Percent |
|---|---|---|
| *Panel A. Perceived diversion from collection* | | |
| Mean collection gap attributed to informal payments | Mean / 4,897 | 62.2 |
| Attribute more than half the gap to informal payments | 3,476 / 4,897 | 71.0 |
| *Panel B. Trust in officials, government and other citizens* | | |
| No trust at all in property-tax staff | 1,831 / 4,897 | 37.4 |
| No trust at all in government | 1,699 / 4,897 | 34.7 |
| No trust at all in other Lahore citizens | 738 / 4,897 | 15.1 |
| *Panel C. Perceived return from taxation* | | |
| Disagree that Punjab uses tax revenue for people like them | 3,147 / 4,897 | 64.3 |

*Notes:* Citizen baseline survey from the same study site, Phase 1. The attribution question allocates the collection gap between informal payments and collection difficulties; its mean is a perceived share, not an audited revenue loss. Trust uses only the unambiguous no-trust endpoint. The service item combines strongly disagree and disagree to some extent. These are separate measures; no index is constructed.

Distrust extends beyond the collection process itself. No trust at all in property-tax staff is reported by 1,831 respondents, or 37.4%, while 1,699 respondents, or 34.7%, report no trust at all in government. By comparison, 15.1% report no trust at all in other Lahore citizens. The pattern therefore points to a particular concern with the fiscal relationship rather than a general unwillingness to trust others. A separate question captures perceptions of the fiscal return: 3,147 respondents, or 64.3%, disagree that the Government of Punjab uses tax revenue to provide goods and services for people like them.

Similarly, a survey of front-line officers provided a complementary view of the same collection environment. Among 292 officials ranking five explanations for incomplete collection, weak penalties and poor public services receive the most first-place rankings. Taxpayers offering informal payments are ranked first by 9.6% of officials for low-value property taxpayers and 8.2% when asked about the medium- and high-value property taxpayers.

## 2.4 Personal Administration and the Move toward Automation

A taxpayer who distrusts the administration may nevertheless value access to an inspector when an assessment is wrong. Property-tax liability depends on recorded property characteristics as well as the valuation rule, so a correctly applied formula can still produce an incorrect bill. Front-line officials can inspect the property, examine supporting documents and initiate authorised corrections.

Such assistance is distinct from the financial advantages available through informal accommodation. Piracha and Moore (2015, 2016) describe how inspectors' local knowledge, continuing relationships with property taxpayers and discretion over assessment and enforcement sustain arrangements that can benefit both parties while limiting public revenue. Personal ad-

ministration can therefore provide access to lawful correction as well as private concessions. I distinguish *non-appeal administrative correction*, an authorised amendment within the assessment chain, from *formal appeal*, which follows the statutory appeal route. A *private concession*, by contrast, reduces the official demand in exchange for a private payment to the inspector. An administrative correction is not evidence of such a bargain.

This combination of assistance and discretion is central to understanding what a less personal administration would replace. Punjab's earlier GIS reforms sought to reduce inspectors' discretion, while the 2024 reform shifted valuation from annual rental value to capital value based on administrative rates (World Bank, 2020; Government of Punjab, 2024). My interviews with senior bureaucrats motivated the experimental comparison between inspector and satellite-based assessment. This comparison is distinct from the implemented valuation reform and does not presume that automation eliminates recourse. Rather, it examines taxpayers' stated choices between assessment arrangements when both the assessment state and the financial terms of personal interaction are explicit. Section 3 formalises these trade-offs in a model that distinguishes preferences over the assessor from willingness to accept a private concession under different assessment states and payment terms.

# 3 The taxpayer's decision environment

Taxpayer choices can depend on how a liability is established and settled, as well as on the amount paid. Research on tax morale identifies nonpecuniary considerations alongside financial incentives (Luttmer and Singhal, 2014). In Punjab, discretion and informal relationships make personal interaction central to the implementation of property taxation (Piracha and Moore, 2015, 2016). I distinguish the financial consequences of a settlement from the initial assessed tax liability, the assessor, and participation in a paid transaction. The framework concerns choices among specified arrangements.

A property taxpayer receives a tax assessment and must decide how to settle the liability. The taxpayer can pay the demand and may have access to an administrative route for correcting an error. An inspector may also offer a lower official payment in return for a bribe. Lawful correction and a paid concession are distinct. The model takes the inspector's assessment and any concession as given and describes the taxpayer's choice among the available terms.

## 3.1 Assessment, payments and payoffs

Let $i$ denote the taxpayer and $j$ a feasible way of settling the liability. The taxpayer sees the initial tax bill, before any correction or concession, and may decide only one of the following three about the bill:

$$e_{ij} \in \{0 \text{ (correct)},\ O \text{ (overstated)},\ U \text{ (understated)}\}.$$

The decision follows the communication of the final payment terms. Total expenditure is

$$\underbrace{p_{ij}}_{\text{total payment}} = \underbrace{T_{ij}}_{\text{final official payment}} + \underbrace{B_{ij}}_{\text{bribe}}, \qquad D_{ij} = \mathbf{1}\{B_{ij} > 0\}.$$

Here $B_{ij} \geq 0$, and $D_{ij}$ indicates whether a bribe is required. Payments and monetary valuations are in rupees. An available lawful correction can change $T_{ij}$ without requiring $B_{ij} > 0$; the process of obtaining it is not separately modelled.

Compare human inspector assessment, $S_{ij} = 0$, with satellite assessment, $S_{ij} = 1$. The satellite arrangement is the prospective automated alternative and has $B_{ij} = D_{ij} = 0$.

## 3.2 Valuations beyond expenditure

A benchmark based only on expenditure, $-p_{ij}$, treats arrangements with equal final total payments as equally attractive. The following terms allow three distinct departures from that benchmark.

**The initial assessment.** The procedural-utility literature distinguishes the value of an outcome from the value of the procedure producing it (Frey et al., 2004). Experiments also distinguish procedural fairness from allocation fairness (Bolton et al., 2005). Motivated by this distinction, let $h_i(e)$ be the valuation of the initial tax bill under human assessment without a bribe, normalized to $h_i(0) = 0$. A taxpayer may prefer receiving a correct bill from the start to receiving an initially excessive bill that is subsequently reduced to the same final payment. This comparison holds the assessor fixed as a human inspector for both choices and excludes informal payments from both offers.

For example, an initial demand of Rs 25,000 that is reduced to Rs 20,000 may be less attractive than a correct initial demand of Rs 20,000, even though both require the taxpayer to pay Rs 20,000. The term $h_i(O)$ captures the monetary value the taxpayer attaches to this difference, relative to $h_i(0) = 0$. Its sign is unrestricted, and it does not separately identify fairness concerns or any other reason for valuing the initial bill.

**The assessor.** Okunogbe and Pouliquen (2022) show that electronic filing can change both the costs of interacting with the tax administration and the financial consequences of that interaction. Personal access can also provide useful services: in a different public-administration setting, Deshpande and Li (2019) show that closing offices providing application assistance reduces applications and receipt of disability benefits. These studies motivate allowing the assessor to matter beyond the displayed payment. Let $\beta_i(e)$ denote taxpayer $i$'s valuation of satellite assessment relative to human inspector assessment when neither offer requires an informal payment. Both profiles have the same initial assessment state $e$ and require the same final tax payment. Thus, $\beta_i(e) > 0$ indicates a preference for satellite assessment, while $\beta_i(e) < 0$ indi-

cates a preference for the human inspector. At $\beta_i(e) = 0$, the taxpayer is indifferent between the assessors on these terms. It can reflect perceived reliability, treatment or access to explanation and correction; it does not separately identify any of them. Its sign is unrestricted, and satellite assessment is not assumed to eliminate human recourse.

**The paid transaction.** Bribery experiments show that external consequences and the interpretation of an exchange can influence behaviour in addition to its private monetary incentives (Barr and Serra, 2009). Banerjee (2016) also identifies a role for entitlement and perceived social appropriateness in strategically equivalent interactions. I therefore distinguish paying $B_{ij}$, which already enters expenditure, from entering a transaction that requires such a payment. Let $m_i(e)$ denote the additional saving required to accept a paid inspector transaction relative to a bribe-free inspector arrangement at the same state. It enters the payoff with a minus sign: $m_i(e) > 0$ denotes reluctance and $m_i(e) < 0$ attraction at equal expenditure. This net valuation is not restricted to a moral cost and is not a second deduction of the bribe amount.

| Symbol | Meaning |
|---|---|
| $h_i(e)$ | Value of the initial state under human assessment without a bribe; $h_i(0) = 0$. |
| $\beta_i(e)$ | Value of automation relative to human assessment, at equal state and payment without a bribe. |
| $m_i(e)$ | Additional transaction cost, beyond the bribe's cash cost in total payment. |

The same payoff function can be written separately for each arrangement:

$$\text{Human inspector:} \quad V_{ij} = \begin{cases} -p_{ij} + h_i(e_{ij}), & D_{ij} = 0, \\ -p_{ij} + h_i(e_{ij}) - m_i(e_{ij}), & D_{ij} = 1; \end{cases} \tag{1}$$

$$\text{Satellite assessment:} \quad V_{ij} = -p_{ij} + h_i(e_{ij}) + \beta_i(e_{ij}), \qquad D_{ij} = 0.$$

The two human cases are settlement without a bribe and a paid concession. Payment utility is locally linear, so the other terms are expressed in monetary units. The additive form is a maintained specification that separates the comparisons of interest; it is not implied by the cited literature. The baseline also restricts the extra transaction valuation to depend on participation and the initial state, rather than separately on the size of a positive bribe. The additional-payment specification in Appendix E examines that restriction.

## 3.3 Choice and acceptance of a concession

Let $\mathcal{J}_i$ be the taxpayer's feasible alternatives. The chosen alternative maximises the payoff:

$$j_i^\star \in \arg\max_{j \in \mathcal{J}_i} V_{ij}.$$

For the same initial state $e$, denote the inspector alternative without a bribe by $H_0$, and the paid concession by $H_1$. The net saving from accepting the concession is

$$s_i \equiv p_{iH_0} - p_{iH_1} = T_{iH_0} - \left(T_{iH_1} + B_{iH_1}\right).$$

Acceptance is optimal relative to the bribe-free alternative when

$$s_i \geq m_i(e). \tag{2}$$

Thus the acceptance threshold, denoted by $s_i^\star(e)$, satisfies

$$s_i^\star(e) = m_i(e), \qquad m_i(O) < m_i(0) \implies s_i^\star(O) < s_i^\star(0).$$

Overassessment lowers the required saving precisely when it reduces the net reluctance to enter the paid transaction. General dislike of an excessive assessment does not by itself produce this change: $h_i(e)$ is common to the two alternatives and cancels.

## 3.4 Testable conditions

Conditions C1–C4 distinguish whether the paid transaction matters beyond expenditure, whether that valuation depends on the initial state, and whether overassessment changes the relative valuation of inspectors. Section 6.1 maps the conditions into empirical restrictions, and Section 6.2 reports the corresponding evidence. Reluctance requires a positive $m_i(e)$; it is not assumed for every taxpayer.

| Economic condition | Parameter restriction | Behavioural implication |
|---|---|---|
| **C1.** Savings alone | $m_i(e) = 0$ for all $e$ | Any positive net saving is enough to prefer a paid concession at the same state. |
| **C2.** Fixed transaction valuation | $m_i(e) = m_i(0)$ | The saving required for acceptance is invariant to the assessment state. |
| **C3.** Overassessment reduces transaction reluctance | $m_i(O) < m_i(0)$ | A smaller net saving is sufficient following overassessment. |
| **C4.** Errors raise the relative value of an inspector | $\beta_i(O) < \beta_i(0)$ | Automation is less attractive relative to human assessment after overassessment, at otherwise comparable terms. |

At equal payment and a common state without a bribe, automation's payoff advantage is $\beta_i(e)$. Relative to that satellite alternative, a paid inspector offer at the same state must therefore save at least $\beta_i(e) + m_i(e)$. Preference for automation can coexist with acceptance of a cheaper

concession. Appendix B derives these comparisons, including alternatives with different initial assessment states.

The literature motivates state dependence without determining its direction. Dufwenberg and Spagnolo (2015) distinguish harassment bribes paid to obtain services to which the payer is entitled, while Banerjee (2016) shows that entitlement can change how a transaction is evaluated. Applied here, an excessive initial demand may make a paid reduction seem more acceptable, giving $m_i(O) < m_i(0)$. This is a hypothesis, not a classification of every concession following overassessment as lawful correction: the final official payment may fall below the correct liability. Likewise, if an error increases the perceived usefulness of personal assistance, $\beta_i(O) < \beta_i(0)$. Evidence on access to administrative services motivates that possibility (Deshpande and Li, 2019); it does not establish the sign for tax assessment.

The two responses are separate from a general dislike of an incorrect initial bill. Comparing paid concessions across states combines $h_i(e)$ with changes in $m_i(e)$; bribe-free alternatives are needed to distinguish them. Trust, inconvenience, moral concerns and perceived correction access can contribute to these net valuations, but the conditions do not identify a unique psychological explanation.

# 4 Data and Procedures

## 4.1 Taxpayers' survey: sampling and recruitment

The taxpayer survey draws on an existing sample of properties with independent remote-sensing measurements. These measurements provide independent information about respondents' own properties. Sampling within this frame therefore links the survey experiment, demand for property evidence and the administrative analysis to the same properties.

The remote-sensing sample comprises 10,564 properties across 48 of Lahore's Excise and Taxation circles, selected from an administrative register of approximately 600,000 properties. Using ESRI satellite imagery from May 2026, the research team independently delineated parcel boundaries, calculated land area from the resulting polygons and predicted covered area using a machine-learning model. These measurements link to administrative records through Property Identification Numbers. They provide a cross-sectional benchmark for the same identified properties, rather than annual measurements of their physical characteristics.

The measurement frame is spatially broad but selective on property type because dense, contiguous and irregular building layouts constrain parcel delineation (Figure A.2). Table 2 reports the resulting coverage differences. Relative to non-selected properties, the remote-sensing sample contains larger plots, greater covered area and higher liabilities; normalised differences for these characteristics range from 0.20 to 0.32. Comparisons with non-selected organised or gated properties are closer, with normalised differences across the reported characteristics of 0.08–0.14. These comparisons describe observable differences in coverage; the survey find-

ings pertain to participating taxpayers within the imagery-covered frame, rather than all Lahore taxpayers.

**Table 2:** Remote-sensing sample and comparison properties

| | RS selected | Organised/ gated | Non-selected | Normalised difference vs org./ gated | Normalised difference vs non-selected |
|---|---|---|---|---|---|
| Land area (sq. yd) | 392.7 | 276.3 | 179.4 | 0.14 | 0.25 |
| Covered area (sq. ft) | 3,497 | 2,723 | 1,736 | 0.09 | 0.20 |
| Tax liability (Rs) | 9,670 | 7,986 | 4,184 | 0.09 | 0.32 |
| Population density | 118.6 | 117.3 | 114.8 | 0.08 | 0.21 |
| Properties | 10,564 | 156,983 | 546,108 | | |

*Notes:* Group means from the project comparison. Organised/gated properties are non-selected properties recorded as organised, planned or gated; they are a subset of all non-selected properties. The normalised difference is $(\bar{x}_{\mathrm{RS}} - \bar{x}_g)/\sqrt{(s^2_{\mathrm{RS}} + s^2_g)/2}$. Density is reported in the source units. The comparison extract and the historical sampling frame differ in coverage.

Using this sampling frame, for the taxpayer survey, interview targets were specified by tax circle, locality and administrative land-area category. Eligible properties were grouped into four land-area bands: 1–5 marla, above 5 through 10 marla, above 10 through 20 marla, and above 20 through 40 marla. The overall target of 2,100 interviews was allocated across localities in proportion to their aggregate recorded covered area. Within each locality, targets were apportioned across land-area-by-storey cells in proportion to their eligible property counts. Properties were randomly ordered within each cell to construct primary and reserve lists. Out of the total sampling frame of 10,564 properties, fieldwork yielded 2,100 completed interviews from 6,088 approached properties. Table 3 summarizes the survey sample.

**Table 3:** Sample definitions and observation flow

| Stage or analysis | Count | Definition |
|---|---|---|
| Remote-sensing sampling frame | 10,564 | 48 circles |
| Approached properties | 6,088 | 46 observed circles; unique property identifiers |
| Completed interviews | 2,100 | 45 observed circles |
| Refusals | 3,988 | All other recorded survey statuses |
| Not approached | 4,476 | Frame less approached properties |
| Substantive information-offer answers | 2,099 | One non-substantive answer |

*Notes:* Property counts refer to unique properties; the experiment has one completed owner interview per property. Frame coverage, survey completion and validity of a particular measurement are different denominators. Valid imagery measurements are reported separately for each attribute and linked analysis. The broader administrative panel contains 23,233 usable property identifiers; all 6,088 approached properties link to it. Earlier baseline citizen and officer surveys are separate sources.

Table 4 compares completed interviews, approached properties without a completed interview, and properties not approached on observed administrative characteristics. It shows that

citizens who completed the surveys are statistically similar to citizens who refused or were not surveyed.

**Table 4:** Survey participation and observed property characteristics

| | (1) | | (2) | | (3) | | Mean differences | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Completed Surveys | | Refused interview | | Not approached | | (1)–(2) | | (1)–(3) | | (2)–(3) | |
| Variable | N | Mean (SD) | N | Mean (SD) | N | Mean (SD) | N | Mean difference | N | Mean difference | N | Mean difference |
| Land Area (marla) | 2100 | 15.361 | 3988 | 16.005 | 4476 | 15.604 | 6088 | -0.644 | 6576 | -0.243 | 8464 | 0.401 |
| | | (16.588) | | (22.870) | | (35.117) | | | | | | |
| Number of floors | 2100 | 2.870 | 3988 | 2.860 | 4476 | 2.535 | 6088 | 0.010 | 6576 | 0.336 | 8464 | 0.325 |
| | | (0.404) | | (0.394) | | (0.680) | | | | | | |
| Total covered area (sqft.) | 2100 | 3072.161 | 3988 | 2758.386 | 4476 | 2924.857 | 6088 | 313.776 | 6576 | 147.304 | 8464 | -166.472 |
| | | (1956.331) | | (1951.123) | | (17482.277) | | | | | | |
| Total property tax | 2100 | 10118.323 | 3988 | 10016.087 | 4476 | 9144.635 | 6088 | 102.235 | 6576 | 973.688 | 8464 | 871.453 |
| | | (12897.526) | | (11196.390) | | (31565.469) | | | | | | |

## 4.2 Survey Measures and Sequence

Fieldwork was conducted between July and August 2026. The consent protocol explained voluntary participation, confidentiality and the hypothetical nature of the choices; participants were offered Rs 1,000 in mobile credit. Appendix C gives survey questions that were asked of the respondents. The survey measures taxpayers' experience with the Excise and Taxation Department, familiarity with their property record, perceptions of recorded land and covered area, and trust in officers' assessments. It distinguishes general beliefs about informal payments from respondents' own reported experiences. Respondents also report recent assessment notices, whether they regard the assessed amount as accurate, awareness of available responses and intentions to seek checking or correction. These measures describe experience, beliefs and intended behaviour; they do not verify that an assessment is legally incorrect or that an application for correction has been submitted.

Documentary resources are measured by whether an taxpayer can produce, within approximately one week, a registry or sale deed, architectural drawings recording land and covered area, and the latest property-tax receipt or PT-1. Before the assessment-choice tasks, respondents report how easy it would be to correct an erroneous computer assessment.

### 4.2.1 Bayesian Adaptive Assessment-Choice Experiment

The survey embedded a choice experiment. Owners chose between receipts displaying the assessor, reference correct bill $Q$, initial bill $I$, informal payment $B$, final official bill $T$ and total expenditure $C = T + B$. The reference anchors the hypothetical task; it is not an independent adjudication of legal liability. Inspector offers included paid concessions while satellite offers did not. Choices neither alter tax records nor implement payments. Figure D.1 reproduces an

example receipt pair; Table D.1 reports the four common comparisons and their eligible samples.

The design varied assessor type, assessment error and concession terms, with expenditure equal to the final official bill plus the informal payment. Four common comparisons precede adaptive questions: assessor choice at equal correct bills; acceptance of a concession between two inspectors; and two comparisons of paid inspector offers differing in initial assessment state. The latter comparisons impose equal displayed expenditure, with the fourth also imposing equal informal payments.

After the four common opening comparisons, question selection adapts to earlier responses. The selection utility separates satellite valuation, transaction valuation and its change under overassessment. This model guides selection and updates respondent preference distributions. The pooled and panel mixed-logit estimates below use a separate ex-post specification, adding initial-error effects, state interactions and position. Appendix E.1 gives the citizen design utility and its reference-bill units.

The satellite offer is always shown first and the human inspector offer second in the opening equal-bill comparison. Inspector offers can contain informal payments; satellite offers do not. Later alternatives depend on earlier answers, so the sequence is not an independently randomised factorial experiment. The analysis contains 21,001 paired tasks and 42,002 offered alternatives from 2,100 completed interviews. Appendix E specifies the likelihood assumptions for this sequential design.

**Table 5:** Common comparisons and their interpretation

| Alternatives / eligible owners | Changes and constants | Comparison and limitation |
|---|---|---|
| 1. Satellite A vs inspector B; $N = 2,100$ (Figure D.2) | Assessor changes. Both initial bills correct, no informal payment, equal total expenditure. | Satellite choice at equal correct bills. Assessor and fixed placement are confounded in this task. |
| 2. Unpaid vs paid inspector; $N = 2,050$ (Figure D.3) | Both inspectors, both initial bills correct. Informal payment, final official bill and total expenditure can change. | Concession acceptance over unpaid inspector offer. No satellite alternative; conditional on displayed eligibility. |
| 3. Paid inspector, correct vs excessive initial bill; $N = 927$ (Figure D.4) | Initial state and informal payment can change. Both inspectors and paid offers; equal total expenditure. | Choice of initial overassessment at equal expenditure. State and payment composition differ; receipt restrictions select the sample. |
| 4. Paid inspector, correct vs excessive initial bill; $N = 1,624$ (Figure D.5) | Initial state changes. Inspector identity, total expenditure and informal payment held constant. | Choice of initial overassessment at equal expenditure and payment. A separate task from row 3; eligibility is not proven independent of respondent characteristics. |
| Later adaptive pairs; tasks five onwards | Eligible inspector and satellite receipts vary in assessor, initial error and financial terms; satellite offers carry no informal payment. | Conditional model-based preferences. History-dependent assignment and model restrictions limit interpretation. |

*Notes:* Counts are the unchanged existing analysis denominators after restrictions on displayed receipts. Monetary amounts depend on the owner's reference liability. The four opening-pair screenshots are in Appendix D; Figure D.1 gives an illustrative respondent-facing receipt. Exact comparison shares appear in Table D.1.

The eligible denominators in Table 5 come from restrictions on the displayed receipts, not interview attrition. Relative to 2,100 opening-task responses, the restrictions omit 50 second-task, 1,173 third-task and 476 fourth-task pairs. The fourth task is a different pair, not a subset of the third. Comparisons use the retained offers as presented.

**Independent evidence offer.** A separate offer concerns computer-estimated details of the taxpayer's property at assigned prices of Rs 0, Rs 100 or Rs 200, deductible from the participation payment. The 2,099 substantive responses provide a behavioural measure for the demand for automated assessment. Appendix C gives the question wording and coding. The offer response does not verify delivery.

## 4.3 Administrative Records and Correction Data

I combine three separate administrative datasets using Property Identification Numbers (PINs): an annual cadastral panel, formal appeal records and PT-1 correction records. These sources provide distinct information on recorded property characteristics, taxpayers' use of the appeal process and corrections documented by the administration.

**Annual cadastral records.** The unbalanced panel contains 23,233 residential properties observed between FY2022–23 and FY2026–27. These properties were drawn from the same administrative cadastre of approximately 600,000 residential properties used to construct the remote-sensing sample. The draw follows the same within-circle stratification by recorded covered area, extended to all 194 tax circles in Lahore, and includes the 10,564 properties forming the taxpayer-survey sampling frame. Annual records report land and covered area, with payable-tax fields available through FY2025–26. The panel therefore allows me to follow changes in recorded property characteristics.

**Formal appeal records.** The appeals dataset covers 2024–2026 and records whether an appeal was made, its filing date, its recorded status and the office handling the case. These fields identify recorded use of the formal appeal route and the administrative status of the resulting cases. Filing an appeal is distinguished from obtaining a correction; the presence of an appeal record alone does not establish that the assessment was amended.

**PT-1 correction records.** A separate dataset contains corrections documented in PT-1 records during 2024–2026. These records contain property characteristics together with comments from the Excise and Taxation Officer (ETO) describing the correction. The comments provide information about the nature of the recorded change that is not available from comparing annual cadastral entries alone. Linking these records to the appeals dataset identifies properties with both a documented correction and a recorded appeal, while retaining the distinction between the two sources. A property-level match does not, by itself, establish that a particular correction resulted from that appeal.

The linked sources distinguish recorded appeals, documented PT-1 corrections and changes in annual property attributes.

# 5 Assessor Choice and Private Concessions

The first two comparisons separate the assessor a taxpayer prefers at equal bills from the financial offer the taxpayer accepts when the terms change. The first holds the bill and payment constant and changes the assessor. The second holds the assessor and initial accuracy constant and offers a lower official bill in exchange for a payment to the inspector. The explicitly correct starting bill separates a private concession from the removal of an excessive assessment.

## 5.1 Assessor preference at equal bills and acceptance of a concession

In the first opening task, each owner chooses between satellite and inspector assessment. Both initial bills are correct, eventual expenditure is the same and neither offer includes an informal payment. Let $Y_i^{\text{sat},1}$ equal one if owner $i$ chooses the satellite and zero if the owner chooses the

inspector. Its mean is the share choosing satellite assessment on these terms, using all 2,100 owners. The satellite is shown first and the inspector second, so this share describes the comparison in its displayed order (Figure D.2).

The second task compares two inspectors, both starting from a bill explicitly described as correct. One retains that bill without an informal payment; the other offers a lower official bill in return for a payment to the inspector. The concession therefore offers an advantage beyond removing an excessive initial demand. Let $A_i$ equal one if the paid offer is chosen and zero otherwise. I retain the 2,050 owners whose two initial bills are correct. Their mean $A_i$ is the concession-acceptance rate. For the joint comparison, I classify these same owners by their first-task choice; the denominator is therefore eligibility for both menus, rather than the full first-task sample (Figure D.3).

The model gives a different decision condition for each menu. At correct initial bills, write $H_0$ for an unpaid inspector, $H_1$ for a paid inspector and $S$ for the satellite. Let $G_i = C_{iH_0} - C_{iH_1}$ be the saving in total expenditure from the paid offer, in rupees. The systematic utility differences are

$$V_{iS} - V_{iH_0} = \beta_i(0), \qquad V_{iH_1} - V_{iH_0} = G_i - m_i(0). \tag{3}$$

Thus satellite assessment is preferred on equal terms when $\beta_i(0) > 0$, while the paid inspector offer is preferred to the unpaid inspector when $G_i > m_i(0)$. Here $m_i(0)$ is the additional saving required to compensate for entering a paid transaction at a correct initial bill. Both inequalities can hold for the same owner. Observed responses also contain choice noise, so an individual acceptance at one saving is not an exact measurement of that owner's threshold.

I report unweighted response shares with Wilson 95% intervals. Panel A of Figure 1 shows that 1,256 of 2,100 owners choose the satellite, or 59.81% (95% interval 57.70–61.89%). The point lies above one half: a majority select satellite assessment when the payment terms are equal. Panel B changes the comparison to paid versus unpaid inspector offers. Here 1,122 of 2,050 owners select the paid concession, or 54.73% (52.57–56.88%). These choices respond to the displayed savings, which average 10.28 reference-bill percentage points and range from 0.0 to 13.33.

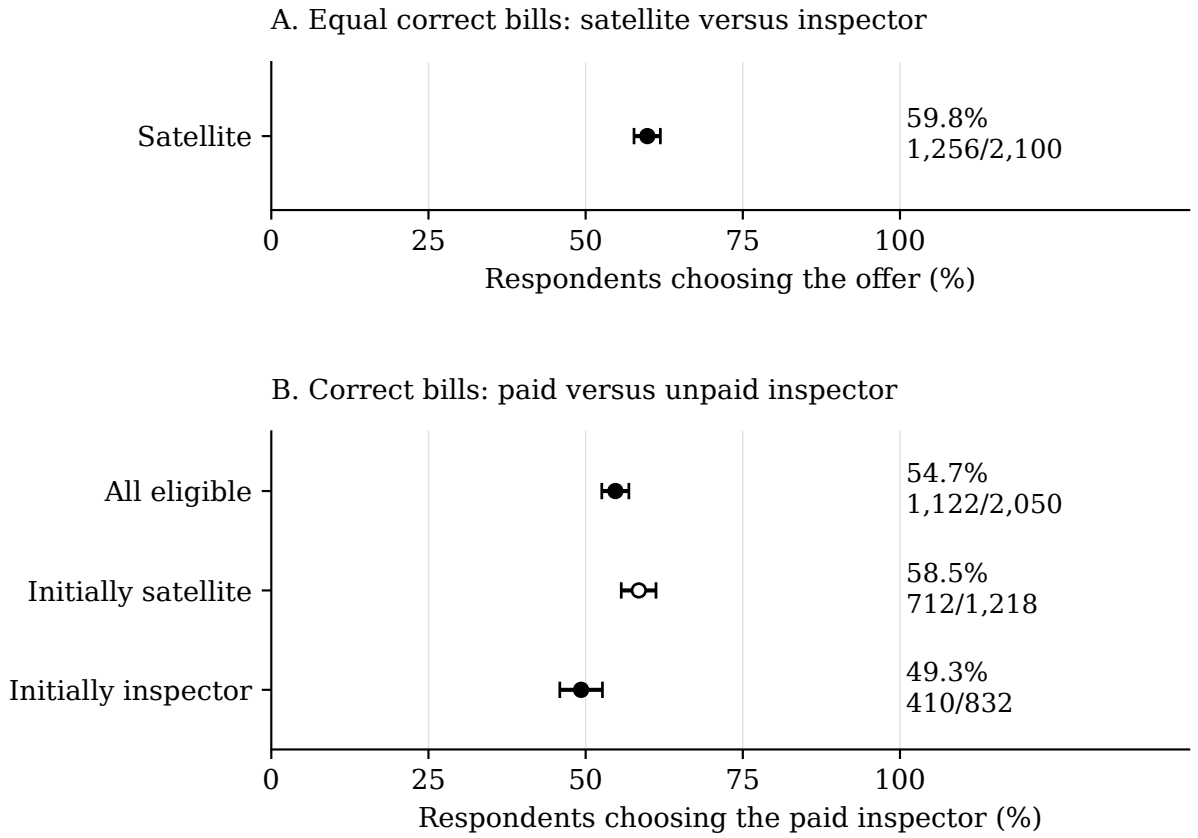



**Figure 1:** Assessor choice and paid-concession acceptance

*Notes:* Panel A: correct initial bills, equal expenditure and no informal payment; satellite is option A and inspector option B. Panel B: two inspectors with correct initial bills, one paid and one unpaid; the paid offer appears as B. Unweighted shares with Wilson 95% intervals. Subgroups condition on Panel A's response. Net savings in the second menu range from 0.0 to 13.33 reference-bill percentage points.

The lower two points in Panel B connect the responses across menus. Among initial satellite choosers, 712 of 1,218 accept the paid inspector offer (58.46%); among initial inspector choosers, 410 of 832 do so (49.28%). The joint satellite-then-paid-inspector group contains 712 owners, or 34.73% of the 2,050 owners eligible for both menus. Preference for satellite assessment therefore coexists with acceptance of a financially attractive inspector concession. These compatible choices concern different menus: the first compares assessors and the second compares inspector offers. The joint response shows acceptance of the concession over an unpaid inspector, with no satellite alternative in that second task.

# 6 Assessment Errors and Transaction Valuations

The opening menus establish that assessor preference and acceptance of a paid concession are different choices. The repeated tasks ask what makes those choices attractive beyond their eventual cost. I distinguish a response to an initially incorrect bill from a change in the value of the assessor or the paid transaction. This distinction is necessary because an owner can dislike an excessive initial demand while becoming more willing to accept a concession that reduces it.

## 6.1 Initial errors at a given total payment

Two common tasks hold expenditure constant while changing the initial assessment. In the third task, both offers are paid inspector arrangements, but one begins with the correct bill and the other with an excessive bill. I retain 927 owners whose displayed expenditures are equal. The outcome is one for selecting the initially overassessed offer and zero for selecting the correct-initial-bill offer. In the fourth task, the informal payments must also be equal, giving a separate sample of 1,624 owners. Figures D.4 and D.5 show the two designs; Appendix Table D.1 reports their response shares and Wilson intervals.

Only 219 of 927 owners (23.6%) select the initially overassessed offer in the expenditure-matched comparison. When the informal payment is also equal, 423 of 1,624 (26.0%) do so. Thus owners usually prefer the arrangement that starts with the correct bill even when it leaves them with the same eventual expenditure. In the model, the systematic utility difference is $h_i(O) - [m_i(O) - m_i(0)]$: it combines dislike of the initial error with any change in transaction valuation. The repeated-choice specification separates these terms by also using unpaid inspector and satellite alternatives.

For owner $i$, task $t$ presents the two offers in $\mathcal{J}_{it}$. The outcome $y_{ijt}$ is one if offer $j$ is selected and zero otherwise. Let $S_{ijt}$ indicate satellite assessment and $D_{ijt}$ a positive informal payment. Relative to reference bill $Q_i$, define initial overassessment by $O_{ijt} = \mathbf{1}\{I_{ijt} > Q_i\}$ and underassessment by $U_{ijt} = \mathbf{1}\{I_{ijt} < Q_i\}$. Total expenditure is $P_{ijt} = (T_{ijt} + B_{ijt})/1000$, measured in thousands of rupees; $\text{Pos}_{ijt}$ equals one for the second displayed receipt. I use the following index to relate the displayed attributes to choice:

$$\begin{aligned} v_{ijt} = {} & \alpha S_{ijt} + b_p P_{ijt} + \kappa D_{ijt} + \gamma_O O_{ijt} + \gamma_U U_{ijt} \\ & + \delta_O S_{ijt} O_{ijt} + \delta_U S_{ijt} U_{ijt} + \lambda_O D_{ijt} O_{ijt} + \lambda_U D_{ijt} U_{ijt} + \pi \text{Pos}_{ijt}. \end{aligned} \tag{4}$$

The reference is an unpaid inspector offer with a correct initial bill. The coefficient $b_p$ describes the response to an additional Rs 1,000 of expenditure. At a correct bill, $\alpha$ compares satellite with inspector assessment, and $\kappa$ compares paid with unpaid inspector offers at the same expenditure. For error state $e \in \{O, U\}$, $\gamma_e$ is the response to the initial error under an unpaid inspector; $\delta_e$ is the change in relative satellite valuation, and $\lambda_e$ is the change in paid-transaction valuation. The coefficient $\pi$ accounts for receipt position.

With type-I extreme-value choice shocks, the probability of choosing offer $j$ is

$$p_{ijt} \equiv \Pr(y_{ijt} = 1 \mid X_{it}) = \frac{\exp(v_{ijt})}{\sum_{k \in \mathcal{J}_{it}} \exp(v_{ikt})}, \qquad j \in \mathcal{J}_{it}, \tag{5}$$

where $X_{it}$ contains the attributes of both offers. A larger index difference makes an offer more likely to be chosen. The coefficients are in logit units: their signs describe the direction of the conditional valuation, while their magnitudes are specific to each fitted model.

Table 6 reports three original estimates. The pooled logit uses all 21,001 paired tasks from 2,100 owners, with standard errors clustered by owner. The panel mixed logit allows persistent owner-specific satellite and paid-transaction coefficients and integrates over each owner's complete sequence; the expenditure coefficient remains common. The third specification restricts the pooled model to tasks five onwards, after the common opening comparisons. Offers in these later tasks depend on prior responses. Conditional interpretation requires assignment to depend on recorded history without additional dependence on current unobserved choice shocks, together with the maintained preference specification. Appendix E gives the likelihood.

**Table 6:** Choices across all valid assessment offers

| Attribute | Pooled logit | Panel mixed logit | Later tasks |
|---|---|---|---|
| Satellite, correct bill | 0.056 (0.047) | 0.074 (0.063) | 0.221 (0.136) |
| Total expenditure, Rs 1,000 | -0.097*** (0.017) | -0.287*** (0.032) | -0.115*** (0.022) |
| Overassessment, unpaid human | -0.625*** (0.089) | -0.971*** (0.109) | -0.245 (0.153) |
| Underassessment, unpaid human | -0.588*** (0.087) | -0.489*** (0.102) | -0.359** (0.148) |
| Satellite × overassessment | 0.093 (0.088) | -0.114 (0.109) | -0.065 (0.159) |
| Satellite × underassessment | 0.142* (0.082) | 0.112 (0.098) | 0.001 (0.154) |
| Paid deal, correct bill | 0.381*** (0.046) | 0.142** (0.063) | 0.316** (0.140) |
| Deal × overassessment | 0.050 (0.084) | 0.289*** (0.103) | 0.146 (0.160) |
| Deal × underassessment | 0.007 (0.083) | 0.281*** (0.099) | 0.073 (0.158) |
| Second display position | -0.244*** (0.019) | -0.302*** (0.023) | -0.126*** (0.021) |
| Tasks | 21,001 | 21,001 | 12,601 |
| Respondents | 2,100 | 2,100 | 2,100 |

*Notes:* Coefficients are in logit units; standard errors in parentheses use respondent clustering or respondent likelihood scores. The mixed model integrates persistent normally distributed satellite and deal coefficients over each owner's complete sequence. Later tasks are slots 5 onward. All specifications use displayed expenditure in thousands of rupees and include the terms shown. No coefficient is labelled a pure moral cost or bargaining parameter. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided normal-reference Wald tests using the reported standard errors. Stars refer to individual coefficients, not joint tests.

These ex-post estimates describe the repeated responses. They are distinct from the citizen engine's fitted $\beta$, $m_0$ and $m_1$, whose monetary coefficient is fixed at $-1$ in reference-bill percentage-point units. Appendix E.1 states that design utility and its units. The original pooled and mixed specifications in Table 6 retain their expenditure measure in Rs 1,000.

Higher expenditure lowers choice in all three specifications. Initially incorrect bills also reduce the attractiveness of an offer after accounting for eventual expenditure. In the pooled model, overassessment and underassessment lower unpaid-inspector utility by 0.625 and 0.588 logit units. For a paid inspector the error contrast is $\gamma_e + \lambda_e$; for satellite assessment it is $\gamma_e + \delta_e$. These sums compare an incorrect with a correct initial bill within the same arrangement.

Figure 2 reports the three contrasts for each error state. A point to the left of zero means that the initial error makes the arrangement less attractive; the horizontal interval records the uncertainty in that comparison, using the full coefficient covariance matrix. The paid-inspector and satellite contrasts are negative as well as the unpaid-inspector contrasts. The later-task overassessment contrast for paid inspectors is less precise and its interval includes zero. The main pattern is therefore a negative valuation of initial errors, even though final expenditure is accounted for. Receiving a correct initial demand has value beyond the amount eventually paid.

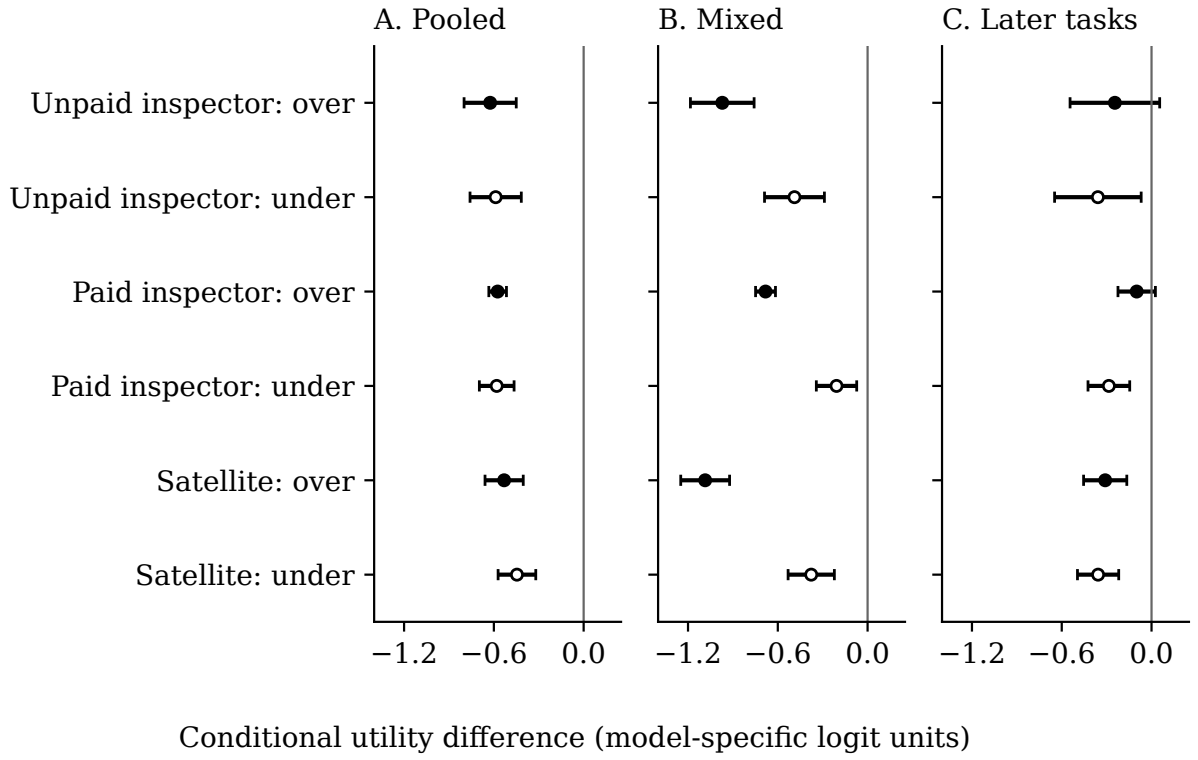



**Figure 2:** Conditional valuation of initially incorrect assessments

*Notes:* Each point compares an initial error with a correct bill for the same assessment arrangement, conditional on total expenditure and position. Horizontal bars are 95% normal-reference intervals using the full coefficient covariance matrix. Separate panels report pooled, panel mixed and later-task models. The zero line denotes no conditional difference; coefficients are in model-specific logit units.

## 6.2 When transaction terms change acceptance

The next comparison asks whether a paid transaction matters once total expenditure and initial accuracy are held fixed. Condition C1 in Section 3.4 sets its additional valuation to zero in every state. In equation (4), this requires $\kappa = \lambda_O = \lambda_U = 0$. I test these restrictions jointly. The pooled and later-task tests concern the common coefficients; the mixed-model test concerns mean transaction valuation. Table 7 rejects the restriction in all three specifications ($p < 0.001$). At a correct initial bill, the paid-transaction coefficient is positive. Through the model's sign convention, this corresponds to negative net reluctance in the pooled model and a negative mean

in the mixed model. The result supports a valuation of the transaction beyond expenditure, rather than a uniformly positive cost of entering it.

Condition C2 asks whether that valuation remains the same when the initial bill is wrong. Its null is $\lambda_O = \lambda_U = 0$. The mixed model rejects this restriction ($p = 0.0011$), whereas the pooled and later-task specifications do not ($p = 0.8361$ and $p = 0.6154$). Direction is a separate question. Under C3, overassessment lowers the saving required for acceptance when $m(O) < m(0)$, corresponding to $\lambda_O > 0$. The mixed estimate is 0.289 with a 95% interval of 0.088–0.491. It therefore favours a lower required saving after overassessment. The other two estimates are positive but imprecise. None of the specifications distinguishes the overassessment shift from the underassessment shift, $\lambda_O - \lambda_U$. The evidence for a change in the acceptance threshold is thus specific to the mixed specification, while the evidence that transaction terms matter is common to all three.

Condition C4 concerns the assessor rather than the transaction. If overassessment raises the relative value of access to an inspector, then $\beta(O) < \beta(0)$ and the satellite interaction $\delta_O$ should be negative. Its estimates have different signs across specifications and all three 95% intervals include zero. The negative error contrasts in Figure 2 therefore give clearer evidence that owners dislike an incorrect initial bill than that overassessment makes inspectors relatively more attractive. Table 7 separates these directional comparisons from the joint tests; Appendix E gives the coefficient mapping and covariance-based calculations.

**Table 7:** Evidence on the model's testable conditions

| Condition or restriction | Pooled | Mixed | Later tasks |
|---|---|---|---|
| *Panel A. Conditions C1–C2: joint-test p-values* | | | |
| C1. Savings alone: $\kappa = \lambda_O = \lambda_U = 0$ | $< 0.001$ | $< 0.001$ | $< 0.001$ |
| C2. Fixed transaction valuation: $\lambda_O = \lambda_U = 0$ | 0.8361 | 0.0011 | 0.6154 |
| *Panel B. Conditions C3–C4: coefficients and 95% intervals* | | | |
| C3. Lower saving threshold: $\lambda_O > 0$ | 0.050 | 0.289 | 0.146 |
| | [-0.114, 0.214] | [0.088, 0.491] | [-0.169, 0.460] |
| C4. Greater inspector value: $\delta_O < 0$ | 0.093 | -0.114 | -0.065 |
| | [-0.079, 0.266] | [-0.327, 0.099] | [-0.376, 0.246] |
| *Panel C. Supplementary restrictions: p-values* | | | |
| State-invariant assessor valuation: $\delta_O = \delta_U = 0$ | 0.1684 | 0.2287 | 0.8161 |
| Equal transaction shifts: $\lambda_O = \lambda_U$ | 0.6865 | 0.9506 | 0.4986 |

*Notes:* Conditions correspond to Section 3.4; their empirical mapping is in Section 6.2. Panels A and C report saved Wald tests using full covariance matrices, with 3, 2, 2 and 1 degrees of freedom, respectively. Panel B reports existing coefficient estimates and two-sided normal-reference intervals; the inequalities state the model predictions, not one-sided $p$-values. The mixed model tests mean transaction valuation; savings alone for every owner would additionally require zero paid-deal heterogeneity. Inference uses respondent clustering or respondent likelihood scores and is unadjusted for multiple comparisons. Nonrejection does not establish equality.

The attribute model also clarifies the comparison absent from the second opening menu. At a common initial state $e$, a paid inspector is preferred in systematic utility to the satellite when the saving relative to the satellite exceeds $m_i(e) + \beta_i(e)$ (Appendix B). The retained estimates organise these expenditure, assessor and transaction terms jointly; the observed joint opening responses alone do not measure that choice. The weak satellite–overassessment interaction is especially relevant here: the repeated menus provide no consistent evidence that an initial excessive bill shifts relative assessor valuation towards inspectors.

The original payment-composition comparison adds a further distinction. Holding total expenditure fixed, an extra rupee paid to the inspector must be offset by one rupee less paid officially. Adding the informal-payment amount gives a coefficient of 0.028 per Rs 1,000 (standard error 0.007; 95% interval 0.014–0.041). This result indicates that the composition of payment contributes to choice as well as the presence of a paid transaction. Expenditure remains negatively valued, both initial-error coefficients remain negative and the satellite–overassessment interaction remains imprecise. The original sensitivity estimates are in Appendix E.

# 7 The Fiscal Allocation of Accepted Concessions

The concession's attractiveness to the taxpayer depends on the saving after paying the inspector. The reduction in official receipts also includes the inspector payment. To make this distinction precise, I use the same correct-initial-bill comparison and require that the unpaid inspector's final bill equals the reference bill $Q_i$. This condition holds for the 2,050 eligible owners. Let 0 denote the unpaid offer and 1 the paid offer. Final official payment is $T_{ij}$, the informal payment is $B_{ij}$ and total expenditure is $C_{ij} = T_{ij} + B_{ij}$. Because $B_{i0} = 0$, the allocation is

$$G_i = C_{i0} - C_{i1}, \qquad B_i = B_{i1}, \qquad R_i = T_{i0} - T_{i1} = G_i + B_i. \tag{6}$$

All three amounts are in rupees. The taxpayer retains $G_i$, the inspector receives $B_i$, and official payment falls by $R_i$. The saving $G_i$ is the amount entering the acceptance condition in equation (3). The official-payment reduction finances this saving and the inspector payment together.

Two denominators answer different questions. The accepted-only calculation describes the allocation among the 1,122 owners selecting the paid offer. The selected-all calculation describes choices among all 2,050 eligible owners, assigning zero saving and zero inspector payment to decliners. A decliner retains the unpaid official bill $Q_i$. With acceptance indicator $A_i$, the selected allocation satisfies

$$\underbrace{Q_i - A_i R_i}_{\text{official payment}} + \underbrace{A_i B_i}_{\text{inspector payment}} + \underbrace{A_i G_i}_{\text{taxpayer saving}} = Q_i.$$

An offered-all average would describe concessions available before selection. The two reported averages instead describe the terms owners select, first among acceptors and then across everyone eligible.

For Figure 3, I divide each receipt amount by its owner's reference bill, multiply by 100 and average equally across owners in the relevant sample. Each bar therefore allocates Rs 100 of reference liability. The dark segment is official payment, the hatched segment is the inspector payment and the unfilled segment is the taxpayer's net saving. Panel A shows that accepted offers leave Rs 76.30 officially, pay Rs 13.30 to the inspector and save the taxpayer Rs 10.40. Official payment falls by 23.70 reference-bill percentage points, more than twice the taxpayer's net saving on this normalised scale.

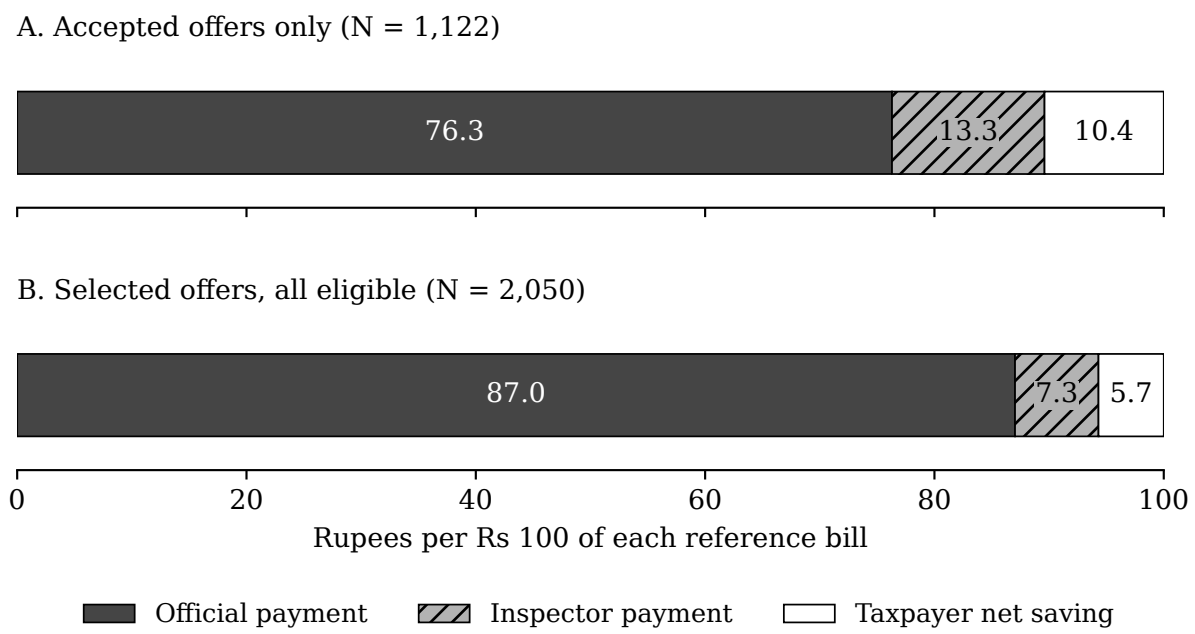



**Figure 3:** Allocation of the reference bill under selected offers

*Notes:* Both initial bills are correct and the unpaid inspector's final bill equals the reference. Each receipt is divided by its own reference bill before averaging, with equal owner weights within a panel. Panel A includes 1,122 acceptors; Panel B all 2,050 eligible owners. Decliners retain official payment of 100 and contribute zero inspector payment and saving. Components sum to 100 subject to rounding. Descriptive accounting of hypothetical choices.

Panel B includes the decliners. Selected official payment is Rs 87.03 per Rs 100 of reference liability, with Rs 7.28 paid to inspectors and Rs 5.69 retained by taxpayers. The official-payment reduction is 12.97 points relative to each owner's unpaid alternative. The smaller reduction reflects the fact that some owners reject the concession and retain the full official bill. This calculation combines the frequency of acceptance with the financial terms selected.

Rupee totals answer a related fiscal question: how much official payment is forgone for each rupee taxpayers retain? I sum the unnormalised selected amounts and divide total official-payment reduction by total taxpayer saving, $\sum_i A_i R_i / \sum_i A_i G_i$. Larger reference bills consequently receive more weight than in the receipt-normalised figure. Table 8 reports Rs 2,732,520

in taxpayer savings, Rs 3,464,400 in inspector payments and Rs 6,196,920 in official-payment reductions. The ratio is 2.27: each rupee saved by taxpayers is accompanied by a little more than two rupees less official payment. The split follows from the terms of the selected hypothetical offers. Acceptance is the behavioural result; the ratio summarises its implication for official receipts.

**Table 8:** Rupee allocation in accepted concessions

| Quantity | Unit | Amount |
|---|---|---|
| Taxpayer net saving, $\sum G_i$ | Rs | 2,732,520 |
| Inspector payments, $\sum B_i$ | Rs | 3,464,400 |
| Reduction in official payment, $\sum R_i$ | Rs | 6,196,920 |
| Official reduction per rupee saved, $\sum R_i / \sum G_i$ | Rs / Rs | 2.27 |

Correct initial bills; unpaid inspector final bill equals the reference. The 1,122 accepted offers are drawn from 2,050 eligible owners. Totals use displayed rupees and therefore give larger bills greater weight than Figure 3, which normalises each receipt first. The ratio divides aggregate amounts. Selected-all totals are identical because decliners contribute zero changes.

Offers beginning with overassessment also separate correction from a further concession. For a selected offer with initial bill $I_i > Q_i$ and lower final bill $T_i$, the reduction $I_i - T_i$ first removes up to $I_i - Q_i$ of initial excess; any amount below $Q_i$ creates a further shortfall, $\max\{Q_i - T_i, 0\}$. Across 4,006 selected adaptive paid offers from 1,699 owners, the average reduction is 32.82 reference-bill percentage points: 10.09 removes initial excess and 22.74 creates a below-reference shortfall. Each selected offer receives equal weight in this supplementary calculation. These selected bundles combine correction of an excessive demand with additional concessions, so their full reduction cannot be read as correction alone.

# 8 Expected Recourse and Assessor Choice

An impersonal assessor may be more attractive when taxpayers expect an accessible way to correct its mistakes. Evidence on property-tax appeals shows that information frictions and unequal assistance from assessors can shape access to redress (Holz et al., 2025). Here the question is how expected recourse relates to the choice of assessor. I examine this relationship first at equal, correct bills, then across the repeated receipt choices. Expected correctability concerns what happens when a computer assessment is wrong. Officer trust concerns whether people in the tax administration treat the owner fairly. I keep the two beliefs separate.

## 8.1 Expected correction at equal bills

Before the choice tasks, S5.5 asks how easy it would be to correct an erroneous computer assessment. I group substantive answers into very difficult, difficult, and easy/very easy. Within

each group, the outcome is the first-task satellite indicator $Y_i^{\text{sat},1}$ defined in Section 5.1. Its mean measures satellite choice when both assessors display the correct bill and the same expenditure. Figure 4 reports unweighted group means and Wilson 95% intervals for the 2,032 owners answering this item.

Satellite choice rises from 41.6% among owners expecting correction to be very difficult to 62.0% among those expecting it to be difficult and 89.6% among those expecting it to be easy or very easy. The horizontal axis is the share choosing the satellite; points further right indicate greater acceptance of impersonal assessment. Because the displayed bills are already correct, the profile concerns expected recourse if a future mistake occurs, rather than correction of an error in the opening receipt.

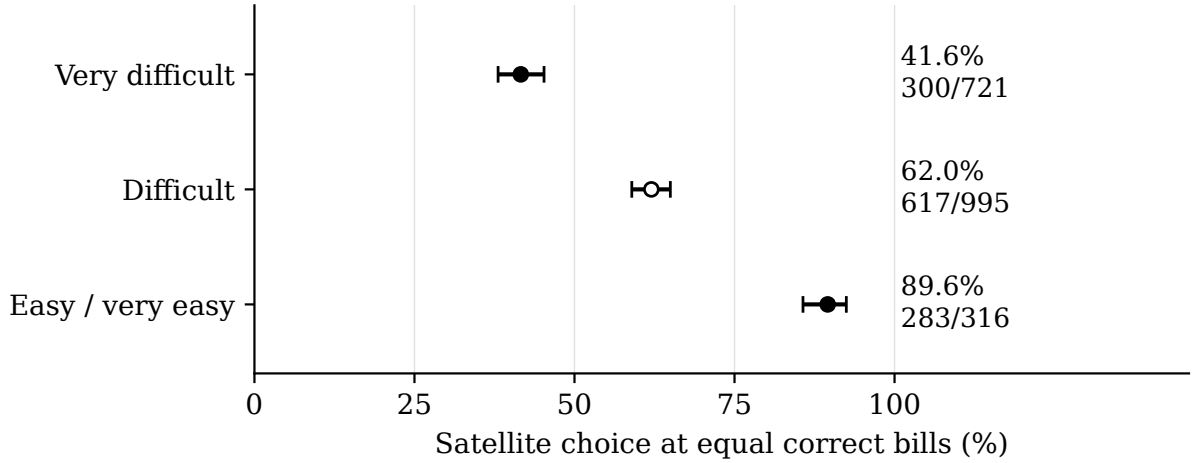



**Figure 4:** Satellite choice by expected ease of correction

*Notes:* First-task satellite choice at equal correct bills among 2,032 substantive S5.5 respondents. Unweighted shares; Wilson 95% intervals. Easy and very easy pool 305 and 11 responses. Counts and denominators are shown. The satellite is first in the opening task. Correctability is elicited after information scripts and before choices.

The groups may also differ in officer trust, beliefs about measurement and the resources available for dealing with the administration. To compare owners with similar measured characteristics, I use the following exploratory secondary regression:

$$100Y_i^{\text{sat},1} = a + \rho E_i + Z_i'\gamma + \eta_{c(i)} + \varepsilon_i. \tag{7}$$

Here $E_i$ is one for easy/very easy correction and zero for difficult/very difficult correction. The outcome is multiplied by 100, so $\rho$ measures the adjusted difference in satellite choice in percentage points. A positive $\rho$ means that expected ease of correction is associated with greater acceptance of the satellite at equal bills. The vector $Z_i$ contains categorical officer trust, relative-mistake beliefs, digital comfort, title status and information arm, plus expected accuracy on its 0–10 scale, document count and log reference bill. Circle fixed effects $\eta_{c(i)}$ compare owners within the same assessment circle; $\gamma$ collects the control coefficients and $\varepsilon_i$ is the unexplained component.

I estimate the equation by ordinary least squares on 1,427 complete cases in 45 circles, clustering standard errors by circle.[1] Correctability is elicited after the information scripts and is not randomly assigned. Unmeasured experience or confidence may affect both beliefs and choice, so these comparisons describe conditional associations rather than the effect of introducing an appeals system. The outcome is observed satellite choice, not the citizen engine's posterior satellite parameter.

The estimated $\rho$ is 10.02 percentage points (95% interval 3.27–16.77; $p = 0.0045$). Owners expecting correction to be easy are therefore about ten percentage points more likely to choose the satellite than owners expecting correction to be difficult, after accounting for the measured differences. The positive interval shows that the association remains when officer trust and perceived measurement performance enter separately. Expected access to correction is consequently relevant to acceptance of an impersonal assessor, alongside beliefs about how well it measures the property.

## 8.2 Recourse and trust across repeated choices

The full sequence asks a related question: does expected correctability remain associated with satellite choice as assessment states and payment terms vary? I enter the beliefs through interactions with the satellite indicator, because an owner-level characteristic common to both receipts otherwise cancels from the utility difference. The most flexible of the original specifications uses

$$v^{\mathrm{C}}_{ijt} = \sum_{r=1}^{3} \mathbf{1}\{W_i = r\} X'_{ijt} \boldsymbol{\theta}_r + S_{ijt}\left(\rho_H H_i + \rho_E E_i + \rho_L L_i\right), \tag{8}$$

with the choice probability in equation (5). The vector $X_{ijt}$ contains the ten receipt terms in equation (4); $\boldsymbol{\theta}_r$ allows their coefficients to differ across the three recorded information groups $W_i$. For this comparison, $H_i$ indicates difficult correction and $E_i$ easy/very easy correction, with very difficult as the reference. The indicator $L_i$ records little or no trust in officers. A positive $\rho_H$ or $\rho_E$ means greater relative valuation of satellite assessment than among owners expecting correction to be very difficult. These coefficients are in logit units, unlike the percentage-point difference in equation (7).

Table 9 builds the comparison in three columns. The first uses common receipt coefficients and correctability interactions for 2,032 owners and 20,320 tasks. The second adds officer trust, leaving 2,004 owners and 20,040 tasks. The third estimates equation (8) on that same sample, allowing receipt valuations to vary by information group. Standard errors are clustered by owner. The positive correctability interactions persist across all three columns: the relationship

[1]The completed secondary analysis uses a common complete-case sample that also requires available citizen-engine parameter means. Those parameters are neither regressors nor outcomes in equation (7).

is present after trust adjustment and after allowing the information groups to value the displayed terms differently.

**Table 9:** Correctability beliefs and full-sequence assessor choices

| Interaction | Beliefs | Add trust | Add arm interactions |
|---|---|---|---|
| Satellite × hard | $0.534^{***}$ (0.071) | $0.561^{***}$ (0.072) | $0.565^{***}$ (0.072) |
| Satellite × easy/very easy | $0.550^{***}$ (0.079) | $0.652^{***}$ (0.082) | $0.648^{***}$ (0.083) |
| Satellite × low officer trust | — | $0.323^{***}$ (0.062) | $0.324^{***}$ (0.063) |
| Respondents | 2,032 | 2,004 | 2,004 |
| Tasks | 20,320 | 20,040 | 20,040 |

*Notes:* All specifications include the terms in equation (4). Very hard is the reference correctability category. Low officer trust combines not at all and a little in the current survey. The third column allows all ten baseline choice coefficients to differ across the three recorded information arms. Correctability is measured after information scripts; these coefficients are descriptive associations. Standard errors are clustered by respondent. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided normal-reference Wald tests using the reported standard errors. Stars refer to individual coefficients, not joint tests.

The adjusted low-trust satellite interaction is 0.324 (standard error 0.063) in the third column. It compares owners reporting little or no trust with those reporting greater trust, conditional on the other terms. First-task shares are not monotonically ordered across all four trust categories, so this coefficient summarises the specified binary contrast. Together, the opening and repeated-choice comparisons distinguish two dimensions of the taxpayer–state relationship: confidence in correcting a computer error and confidence in fair treatment by an officer.

# 9 Demand for Independent Property Information

Independent property information can help an owner understand a record or question a bill under either assessor. The information offer therefore measures a different demand from the choice of who assesses the property. S6.1 offers computer-estimated details of the owner's own property for Rs 0, Rs 100 or Rs 200, with any price deducted from participation credit. Let $\text{Buy}_i$ equal one for acceptance and zero for rejection, and let $q_i$ be the recorded offer price in rupees. I use the 2,099 substantive responses.

I compare each paid offer with free provision using

$$100\text{Buy}_i = \alpha_B + \tau_{100}\mathbf{1}\{q_i = 100\} + \tau_{200}\mathbf{1}\{q_i = 200\} + \varepsilon_i. \tag{9}$$

The intercept $\alpha_B$ is acceptance at the free offer, in percentage points. Each $\tau$ is the difference at the stated price relative to free provision. A negative coefficient means lower acceptance at that price. This saturated specification estimates the two differences separately rather than imposing a linear response to price. HC3 standard errors allow different variances across groups. The estimates describe differences across the recorded price groups.

Figure 5 reports acceptance of 94.8% when free, 39.0% at Rs 100 and 39.4% at Rs 200. The largest difference is between free provision and either positive price. The fitted differences are −55.7 and −55.4 percentage points, each with an HC3 standard error of approximately two points (Table F.1). Acceptance is similar at the two positive prices. Independent information attracts broad interest when free, while a substantial share of owners also accept it when payment is required.

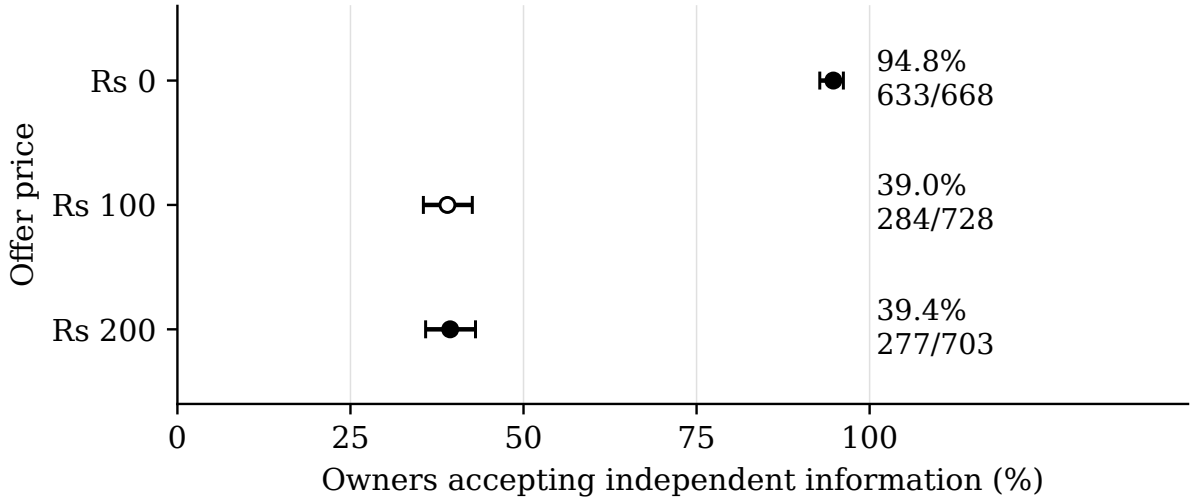



**Figure 5:** Acceptance of independent property information by offer price

*Notes:* S6.1 offers computer-estimated details of the owner's property, with price deducted from participation credit. There are 2,099 substantive responses; one additional owner at Rs 200 provides no substantive answer. Unweighted shares and Wilson 95% intervals; counts and denominators appear beside the points. The outcome records offer acceptance.

The price comparison concerns demand for information about the owner's own property. The earlier information scripts instead provide a satellite explanation and, in T2, an estimate for a comparable property. I compare these recorded groups using unadjusted differences with HC1 standard errors and item-specific eligible samples. Expected recourse is coded one for easy/very easy, the choice and acceptance outcomes are binary, and expected accuracy retains its 0–10 count scale. Assignment is not independently verified in the saved analysis, so I report these as exploratory group comparisons in Table F.2. The completed belief and opening-choice contrasts show no clear average changes: the intervals include zero and retain economically relevant differences in both directions.

The economic distinction is between information that helps a taxpayer understand the record and the institution that determines the bill. The demand result supports the relevance of independent property evidence, while the recourse result connects acceptance of impersonal assessment to expected responsiveness when something goes wrong. The correction-readiness evidence in Section 2 places both choices in a setting where obtaining documents and pursuing a correction can require effort.

# 10 Conclusion

Preference for an impersonal assessor coexists with acceptance of inspector-mediated private concessions. The two opening menus establish this distinction directly: many owners choose satellite assessment at equal correct bills, and many of those owners subsequently accept a paid reduction when choosing between two inspectors. Since the initial bill is explicitly correct, the concession provides a financial advantage beyond removing an excessive demand. Its terms reduce official payment by more than the taxpayer's net saving because part of the reduction finances the inspector payment.

The original repeated-choice estimates show that initial errors and the terms of a paid transaction matter beyond eventual expenditure. Expected ease of correcting a computer error remains associated with satellite choice after accounting for officer trust, measurement beliefs and resources. Demand for independent property information adds a related finding: owners value evidence that can help them understand a record under either assessor. These results place access to explanation and correction alongside private financial incentives in the taxpayer–state relationship.

For public finance, citizen acceptance is one dimension of an assessment institution. Equal treatment and official receipts depend on the terms being accepted, while usable recourse concerns how taxpayers can challenge mistakes. The evidence therefore distinguishes the service provided by frontline officers from the concessions their discretion can facilitate. Assessing impersonal administration requires considering both the accuracy of its demands and the accessibility of correction, alongside the allocation of the tax claim.

# References


**Autor, David H, Frank Levy, and Richard J Murnane**, "The skill content of recent technological change: An empirical exploration," *The Quarterly journal of economics*, 2003, *118* (4), 1279–1333.

**Banerjee, Ritwik**, "On the Interpretation of Bribery in a Laboratory Corruption Game: Moral Frames and Social Norms," *Experimental Economics*, 2016, *19* (1), 240–267.

**Barr, Abigail and Danila Serra**, "The Effects of Externalities and Framing on Bribery in a Petty Corruption Experiment," *Experimental Economics*, 2009, *12* (4), 488–503.

**Besley, Timothy and Torsten Persson**, "Why Do Developing Countries Tax So Little?," *Journal of Economic Perspectives*, 2014, *28* (4), 99–120.

**Bolton, Gary E., Jordi Brandts, and Axel Ockenfels**, "Fair Procedures: Evidence from Games Involving Lotteries," *The Economic Journal*, 2005, *115* (506), 1054–1076.

**Bovens, Mark and Stavros Zouridis**, "From street-level to system-level bureaucracies: How information and communication technology is transforming administrative discretion and constitutional control," *Public administration review*, 2002, *62* (2), 174–184.

**Bullock, Justin B**, "Artificial intelligence, discretion, and bureaucracy," *The American Review of Public Administration*, 2019, *49* (7), 751–761.

**Deshpande, Manasi and Yue Li**, "Who Is Screened Out? Application Costs and the Targeting of Disability Programs," *American Economic Journal: Economic Policy*, 2019, *11* (4), 213–248.

**Dufwenberg, Martin and Giancarlo Spagnolo**, "Legalizing Bribe Giving," *Economic Inquiry*, 2015, *53* (2), 836–853.

**Frey, Bruno S., Matthias Benz, and Alois Stutzer**, "Introducing Procedural Utility: Not Only What, but Also How Matters," *Journal of Institutional and Theoretical Economics*, 2004, *160* (3), 377–401.

**Gangl, Katharina, Eva Hofmann, and Erich Kirchler**, "Tax Authorities' Interaction with Taxpayers: A Conception of Compliance in Social Dilemmas by Power and Trust," *New Ideas in Psychology*, 2015, *37*, 13–23.

**Government of Punjab**, "The Punjab Finance Act 2024," Excise, Taxation and Narcotics Control Department, Government of Punjab 2024. Punjab Act XIII of 2024.

**Gräfe, Philipp, Justine Marienfeldt, Liz Marla Wehmeier, and Sabine Kuhlmann**, "Changing tasks and changing public servants? The digitalisation and automation of public administrative work," *Information Polity*, 2024, p. 13837605241289773.

**Holz, Justin, David Novgorodsky, and Andrew Simon**, "Disparities in Tax Negotiations: Evidence from Field Experiments with Homeowners and Assessors," 2025.

**Khan, Adnan Q., Asim Ijaz Khwaja, and Benjamin A. Olken**, "Tax Farming Redux: Experimental Evidence on Performance Pay for Tax Collectors," *The Quarterly Journal of Economics*, 2016, *131* (1), 219–271.

**Kirchler, Erich, Erik Hoelzl, and Ingrid Wahl**, "Enforced versus Voluntary Tax Compliance: The "Slippery Slope" Framework," *Journal of Economic Psychology*, 2008, *29* (2), 210–225.

**Knebelmann, Justine, Victor Pouliquen, and Bassirou Sarr**, "Discretion versus Algorithms: Bureaucrats, Tax Equity and Acceptability," Working Paper 26/34, Institute for Fiscal Studies 2026.

**Luttmer, Erzo F. P. and Monica Singhal**, "Tax Morale," *Journal of Economic Perspectives*, 2014, *28* (4), 149–168.

**Mascagni, Giulia, Celeste Scarpini, Denis Mukama, Fabrizio Santoro, and Naphtal Hakizimana**, "The Interpersonal Side of Tax Compliance: Interactions Between Taxpayers and Tax Officials in Rwanda," ICTD Working Paper 215, Institute of Development Studies 2025.

**Murphy, Kristina**, "The Role of Trust in Nurturing Compliance: A Study of Accused Tax Avoiders," *Law and Human Behavior*, 2004, *28* (2), 187–209.

**Okunogbe, Oyebola and Victor Pouliquen**, "Technology, Taxation, and Corruption: Evidence from the Introduction of Electronic Tax Filing," *American Economic Journal: Economic Policy*, 2022, *14* (1), 341–372.

**Piracha, Mujtaba and Mick Moore**, "Understanding Low-Level State Capacity: Property Tax Collection in Pakistan," ICTD Working Paper 33, International Centre for Tax and Development, Brighton 2015.

__ **and** __ , "Revenue-Maximising or Revenue-Sacrificing Government? Property Tax in Pakistan," *Journal of Development Studies*, 2016, *52* (12), 1776–1790.

**Slemrod, Joel**, "Tax Compliance and Enforcement," *Journal of Economic Literature*, 2019, *57* (4), 904–954.

**van Dijke, Marius and Peter Verboon**, "Trust in Authorities as a Boundary Condition to Procedural Fairness Effects on Tax Compliance," *Journal of Economic Psychology*, 2010, *31* (1), 80–91.

**World Bank**, "Property Tax Diagnostic Manual," Technical Report, World Bank, Washington, DC 2020. Case study on Punjab Province, Pakistan.

# A Additional Figures

## A.1 Administrative organisation

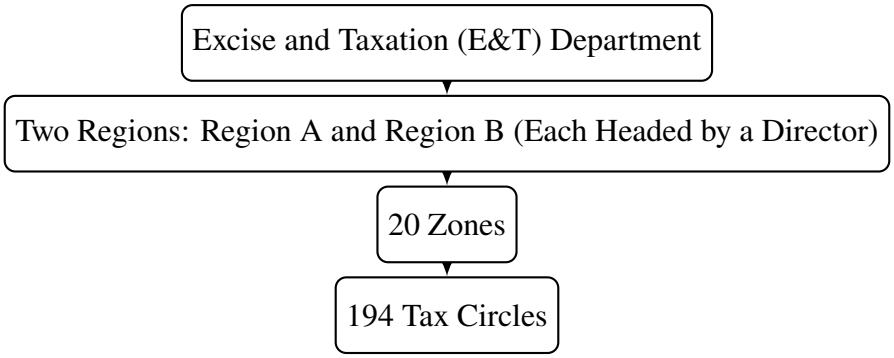



**Figure A.1:** Flowchart of Property Tax Administration in Lahore

## A.2 Spatial coverage for survey sampling frame

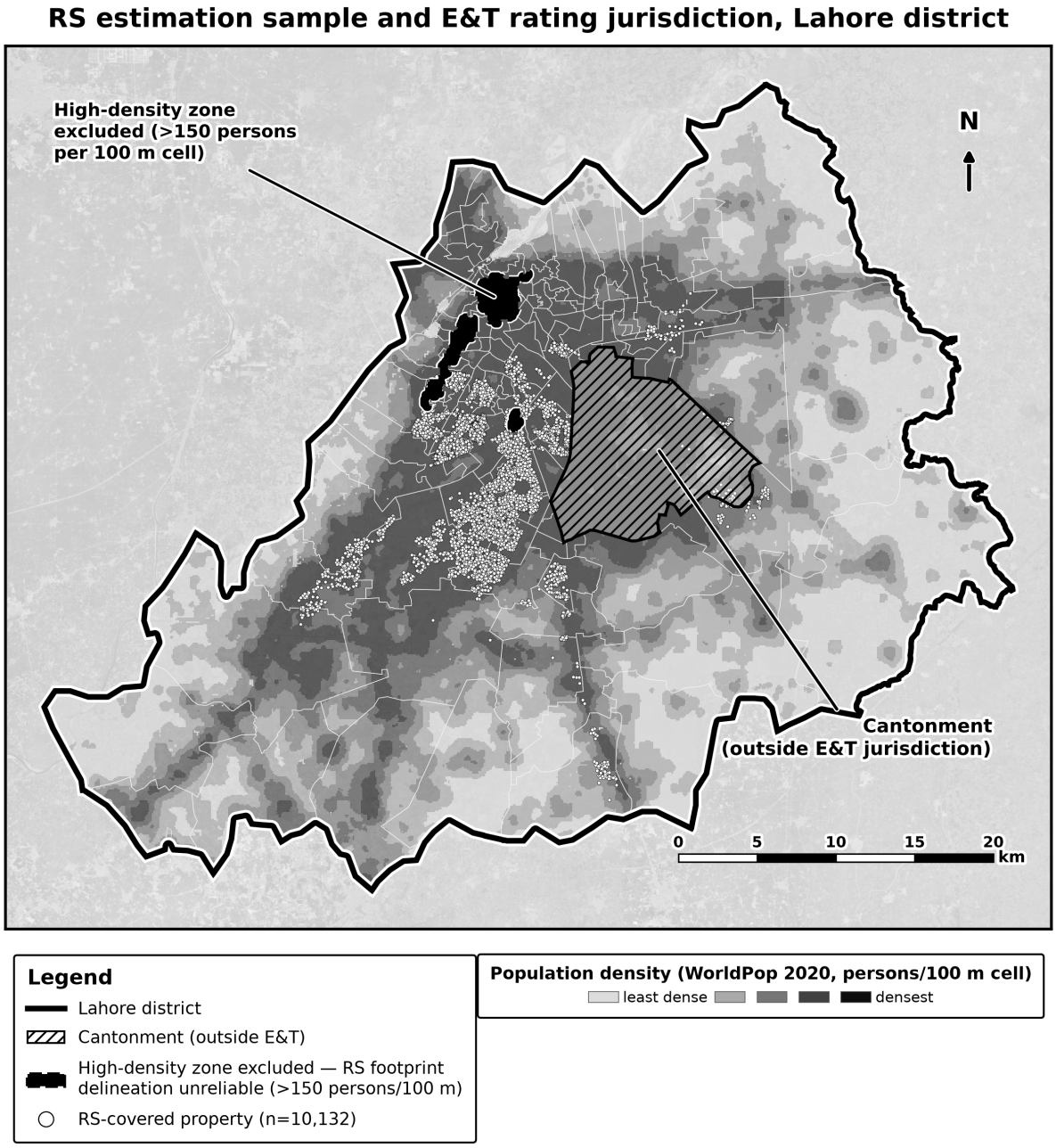



**Figure A.2:** The remote-sensing estimation sample within Lahore district. Grayscale shading is the WorldPop 2020 population surface (denser cells darker); thin lines are E&T rating jurisdiction. The Lahore Cantonment (hatched) is rated by the Cantonment Board and lies outside E&T jurisdiction; the dashed zone marks the high-density old-city fabric (above 150 persons per 100 m cell) where footprint delineation from imagery is unreliable. Open circles are the $n$ = 10,564 RS-covered properties.

# B Payoff comparisons and acceptance conditions

## B.1 Utility comparisons

Write $H_0$ for an inspector alternative without a bribe, $H_1$ for a paid inspector alternative, and $S$ for an automated alternative. For inspector alternatives at the same initial state $e$, define the net saving $s = p_{H_0} - p_{H_1}$. Suppressing the owner subscript, equation (1) gives

$$\begin{aligned} V_{H_1} - V_{H_0} &= [-p_{H_1} + h(e) - m(e)] - [-p_{H_0} + h(e)] \\ &= s - m(e). \end{aligned}$$

Thus $V_{H_1} \geq V_{H_0}$ if and only if $s \geq m(e)$, proving equation (2), including zero and negative $m(e)$.

For alternatives without bribes with common payment and state,

$$V_S - V_{H_0} = [-p + h(e) + \beta(e)] - [-p + h(e)] = \beta(e).$$

Comparing this utility difference at $e$ and 0 gives $\beta(e) - \beta(0)$. Direct state valuation cancels within each comparison.

Likewise, define the paid-minus-bribe-free inspector contrast at common expenditure as $\Delta_D(e) = V_{H_1}(p, e) - V_{H_0}(p, e)$. Then

$$\Delta_D(e) = -m(e), \qquad \Delta_D(e) - \Delta_D(0) = -[m(e) - m(0)].$$

This isolates state dependence in the paid-transaction term under the model. By contrast, two paid inspector alternatives at equal expenditure give

$$V_{H_1}(p, e) - V_{H_1}(p, 0) = h(e) - [m(e) - m(0)],$$

using $h(0) = 0$. The general error response remains. Equal bribe amounts do not remove it.

Finally, compare a paid inspector alternative at state $e_H$ with automation at state $e_S$. Defining $s = p_S - p_H$,

$$\begin{aligned} V_H - V_S &= [-p_H + h(e_H) - m(e_H)] - [-p_S + h(e_S) + \beta(e_S)] \\ &= s + h(e_H) - h(e_S) - m(e_H) - \beta(e_S). \end{aligned}$$

The inspector offer is weakly preferred precisely when

$$s \geq \beta(e_S) + m(e_H) + h(e_S) - h(e_H).$$

At a common state this reduces to $\beta(e) + m(e)$. Omitting the state-valuation difference when states differ requires an additional restriction.

# C Survey questions and coding

The following extract retains the questions required to interpret the reported survey measures. Each non-BACE outcome has its own coding and denominator. Codes 97 (don't know) and 98 (refusal), blanks and other non-substantive values are excluded from substantive-response percentages and shown separately where relevant. The 2,100 completed owner interviews are distinct from the contextual Phase 1 citizen and bureaucrat surveys.

**Table C.1:** Questionnaire items for the outcomes and respondent characteristics

| Measure and item | Question | Responses and routing |
|---|---|---|
| Expected correctability S5.5 | If a computer assessment of your property were wrong, how easy to get it corrected? | 1 = Very hard; 2 = Hard; 3 = Easy; 4 = Very easy. Asked before the choice tasks, after earlier survey material. |
| Evidence demand S6.1 | You can receive the computer's estimated details for your own house (covered area, plot outline). The price would be taken from your Rs 2,000 survey payment. Would you like to buy this for Rs ⟨0/100/200⟩? (Rs 0 = free) | Yes; No; Don't know/undecided; Refused. Offered after the choice tasks; the recorded prices are Rs 0, Rs 100 and Rs 200. |
| Correction intention S9.1 | In the next 3 months, how likely are you to apply to check or correct this house's tax record – out of 10? | 0–10. Non-substantive responses use separate side codes. |
| Reason for inaction S9.2 | If you do NOT plan to contest, what is the main reason? | 1 = Assessment seems fair; 2 = Too costly to appeal; 3 = Don't know how to appeal; 4 = Afraid of consequences; 5 = Too much hassle; 77 = Other (specify). Asked when intention is 0–4. |
| Digital comfort S1.6 | How comfortable are you with filling a form online? | 1 = Not at all comfortable; 2 = A little comfortable; 3 = Comfortable; 4 = Very comfortable. |
| Document access S3.17 | When you need them for property or other official work, how easy or difficult is it to find these documents for this property? | 1 = Very difficult; 2 = Difficult; 3 = Somewhat difficult; 4 = Easy; 5 = Very easy. |

**Table C.1:** Questionnaire items for the outcomes and respondent characteristics (continued)

| Measure and item | Question | Responses and routing |
|---|---|---|
| Officer trust S3.9 | In general, how much do you trust property-tax officers to value properties fairly? | 1 = Not at all; 2 = A little; 3 = Mostly; 4 = Completely. |
| Recent reassessment S3.6 | In last six months, have you received any communication from E&T about change in assessment either informally or formally through a letter | 1 = Yes; 0 = No. |
| Notice accuracy S3.7 | If yes, is the new assessed value accurate? | 1 = Much too low; 2 = Somewhat too low; 3 = About right; 4 = Somewhat too high; 5 = Much too high. Asked following a reported reassessment notice. |

*Notes:* Questions use the fielded instrument and its English extracts, with typographic standardisation. The evidence-offer row uses the fielded English labels with the observed price support inserted. Except for the evidence offer, the listed items additionally allow 97 = Don't know and 98 = Refused (or Refused to answer). These responses remain outside the substantive scales. Correctability combines Easy and Very easy in the analysis and is a pre-choice belief, not an assigned treatment or a measure established to precede every survey exposure. The fielded offer price is populated from the respondent's preloaded record. Acceptance is distinguished from verified payment, delivery and use in a successful correction. The analysis codes evidence acceptance as one and rejection as zero; other responses are excluded. The archived English S3.6 wording specifies six months, whereas its Urdu version specifies one year; the text therefore refers to a recent communication. The fielded form places officer trust before the information scripts, expected correctability after the scripts and before the choice tasks, and the evidence offer after the tasks.

**Table C.2:** Contact with the tax administration: separate earlier survey

| Reported contact or reason | Count / eligible | Percent | 95% interval |
|---|---|---|---|
| Any household contact in last 12 months | 157/399 | 39.3 | [34.7, 44.2] |
| Paying tax bill | 72/157 | 45.9 | [38.3, 53.7] |
| Correcting property records | 16/157 | 10.2 | [6.4, 15.9] |
| Registering new property | 8/157 | 5.1 | [2.6, 9.7] |
| Responding to assessment notice | 3/157 | 1.9 | [0.7, 5.5] |
| General information/inquiry | 72/157 | 45.9 | [38.3, 53.7] |
| Refused | 4/157 | 2.5 | [1.0, 6.4] |
| Do not know | 2/157 | 1.3 | [0.4, 4.5] |
| Other | 0/157 | 0.0 | [-0.0, 2.4] |

*Notes:* The March–April 2026 survey contributes 399 completed interviews (`status_survey=1`). S7.1a asks whether the respondent or anyone in the household had contact with E&T regarding the property in the last 12 months. S7.1d allows multiple reasons among the 157 reporting contact; percentages therefore need not sum to 100. All contact respondents remain in the denominator, including refusals and unknown answers. Intervals are pointwise Wilson intervals under independent sampling, provided to indicate finite-sample uncertainty rather than population representativeness. The earlier sample is not pooled with the 2,100-owner experiment. Contact reasons do not establish what an official did or whether a correction succeeded.

# D BACE conjoint: respondent-facing scenarios

Figure D.1 and Figures D.2–D.5 reproduce the five original illustrations supplied with the design documentation: an example receipt pair and the four common opening scenarios. The images retain their original wording, option order and displayed amounts. The Rs 20,000 reference is illustrative; it is not a common liability assigned to every owner. Sections 5.1 and 6.1 define the common-task outcomes and explain the repeated-choice estimation. Table D.1 records the realised eligible samples, which differ from these illustrative screens.




**Figure D.1:** Scenario 1: satellite and inspector at equal correct bills

| | OPTION A | OPTION B |
|---|---|---|
| **Your real bill** | **Rs 20,000** | **Rs 20,000** |
| **Who checks your house** | A satellite checks automatically | An inspector visits & measures |
| **Measuring mistake (this time)** | **Correct amount (no mistake)** | **Correct amount (no mistake)** |
| **Bill after the mistake** | **Rs 20,000** | **Rs 20,000** |
| **The deal (chai-pani)** | **No deal, no one to bargain with** | **No payment offered** |
| **Informal payment** | **Rs 0** | **Rs 0** |
| **Final recorded bill** | **Rs 20,000** | **Rs 20,000** |
| **What actually leaves your pocket** | **Rs 20,000** | **Rs 20,000** |

Choose A Choose B

**Figure D.2:** Scenario 1: satellite and inspector at equal correct bills

| | OPTION A | OPTION B |
|---|---|---|
| **Your real bill** | Rs 20,000 | Rs 20,000 |
| **Who checks your house** | An inspector visits & measures | An inspector visits & measures |
| **Measuring mistake (this time)** | Correct amount (no mistake) | Correct amount (no mistake) |
| **Bill after the mistake** | Rs 20,000 | Rs 20,000 |
| **The deal (chai-pani)** | No payment offered | Pay Rs 3,000 and the inspector records Rs 14,600 instead of Rs 20,000 |
| **Informal payment** | Rs 0 | Rs 3,000 |
| **Final recorded bill** | Rs 20,000 | Rs 14,600 |
| **What actually leaves your pocket** | **Rs 20,000** | Rs 14,600 + Rs 3,000<br>**Rs 17,600** |

Choose A | Choose B

**Figure D.3:** Scenario 2: inspector offers with and without a paid concession

| | OPTION A | OPTION B |
|---|---|---|
| **Your real bill** | **Rs 20,000** | **Rs 20,000** |
| **Who checks your house** | An inspector visits & measures | An inspector visits & measures |
| **Measuring mistake (this time)** | **Correct amount (no mistake)** | **+ Rs 3,000 recorded too HIGH (by mistake)** |
| **Bill after the mistake** | **Rs 20,000** | **Rs 23,000** |
| **The deal (chai-pani)** | **Pay Rs 2,000 and the inspector records Rs 17,000 instead of Rs 20,000** | **Pay Rs 4,000 and the inspector records Rs 15,000 instead of Rs 23,000** |
| **Informal payment** | **Rs 2,000** | **Rs 4,000** |
| **Final recorded bill** | **Rs 17,000** | **Rs 15,000** |
| **What actually leaves your pocket** | Rs 17,000 + Rs 2,000 **Rs 19,000** | Rs 15,000 + Rs 4,000 **Rs 19,000** |

Choose A | Choose B

**Figure D.4:** Scenario 3: different initial assessment states at equal total expenditure

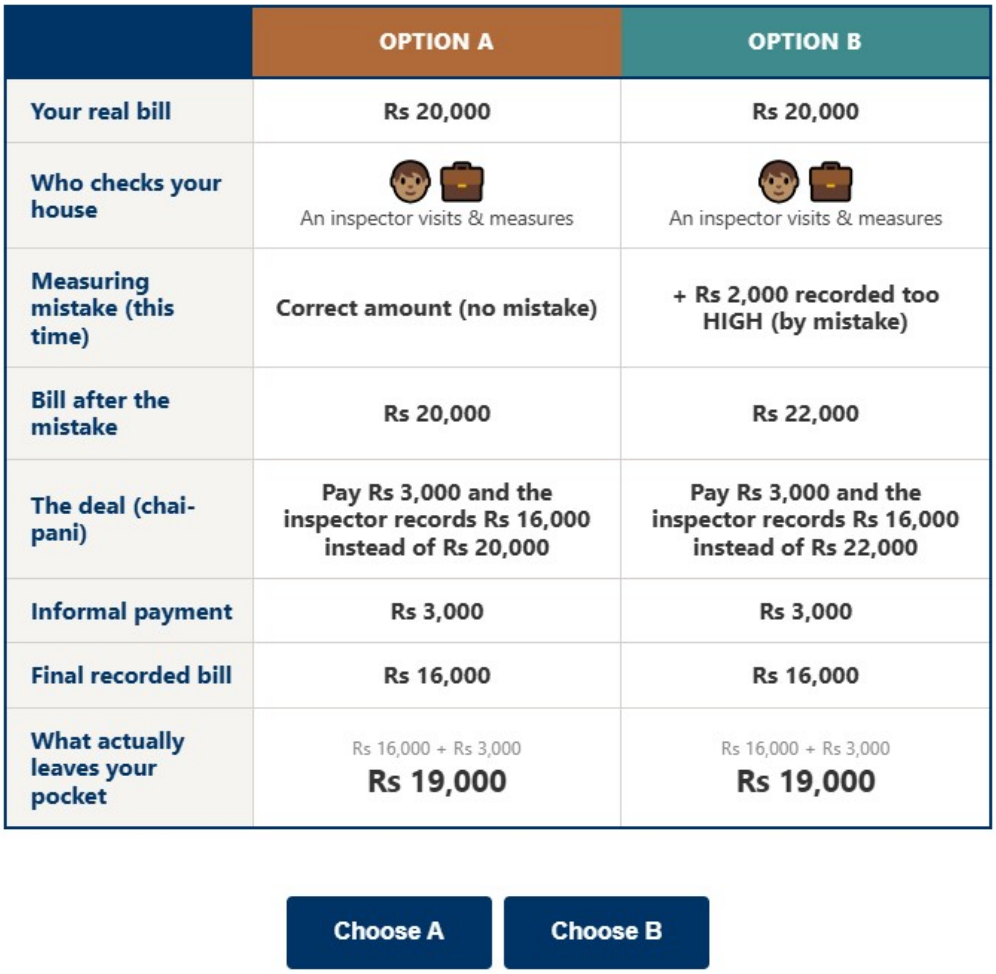



**Figure D.5:** Scenario 4: different initial assessment states at equal expenditure and informal payment

Table D.1 applies each comparison's restrictions to the displayed receipt amounts. Failing an equality condition excludes a task from that common-comparison share, not automatically from full-sequence estimation. The reference correct bill is part of the hypothetical exercise, not an independent adjudication of legal liability.

**Table D.1:** Common comparisons satisfying the displayed conditions

| Choice | Chosen / eligible | Percent | 95% interval |
|---|---|---|---|
| Satellite: correct bill, equal expenditure, no deal | 1,256/2,100 | 59.8 | [57.7, 61.9] |
| Inspector deal: correct bills, deal versus no deal | 1,122/2,050 | 54.7 | [52.6, 56.9] |
| Overassessment-starting deal: equal expenditure | 219/927 | 23.6 | [21.0, 26.5] |
| Overassessment-starting deal: equal expenditure and payment | 423/1,624 | 26.0 | [24.0, 28.2] |

*Notes:* Each initial task has 2,100 choices. Eligibility imposes the stated conditions on the receipts. Both sides of tasks 2–4 are inspectors; tasks 3–4 contain payments on both sides. Median net savings in the qualifying second task are Rs 1,200. Wilson intervals describe choices as presented, without removing position or order effects.

# E Sample description, likelihood and sensitivity

The administrative comparison links unique property identifiers rather than treating each annual record as a separate property. The survey file contains 6,088 unique properties, all of which match the linked administrative panel; 2,100 have a completed interview. The size comparison uses positive observed values and reports the denominator separately for each source and participation group.

The survey's trust and corruption items distinguish institutional perceptions from personal experience. Among valid responses, 1,054 of 2,072 owners report little or no trust that officers value properties fairly. Of 2,034 answering the general informal-payment prevalence question, 102 report often or very often, and 535 report sometimes or more frequently. Among 1,188 contacted owners giving a valid personal-experience answer, 159 report being asked for an informal payment. These different quantities neither replicate the baseline collection-gap attribution question nor form a time series with it.

**Table E.1:** Support across the complete choice sequence

| Assessor | Initial bill | No paid deal | Paid deal |
|---|---|---|---|
| Inspector | Correct | 4,735 | 7,754 |
| Inspector | Overassessed | 886 | 10,815 |
| Inspector | Underassessed | 1,097 | 3,702 |
| Satellite | Correct | 4,071 | 0 |
| Satellite | Overassessed | 3,869 | 0 |
| Satellite | Underassessed | 5,073 | 0 |

*Notes:* Counts are offered alternatives, not respondents or chosen offers. The 42,002 alternatives form 21,001 paired tasks from 2,100 owners. Satellite offers never include a paid deal. The analysis uses displayed rupee amounts and preserves each original task slot.

Receipt attributes are reconstructed from the amounts shown. Error equals initial minus correct liability; the official reduction equals initial minus final official liability; expenditure equals the final official liability plus the requested payment. Every retained receipt satisfies these identities and each parsed paid-offer sentence agrees with the stated amounts. Four alternatives show a Rs 100 reduction without an informal payment; the paid-deal indicator requires a strictly positive payment. Every respondent supplies ten valid tasks. One also supplies a valid slot-12 task; the intervening slot has no usable choice. Omitting the additional task leaves the substantive results unchanged.

For two offers $j, k \in \mathcal{J}_{it}$, conditioning on the task eliminates any term common to the pair. Equation (5) implies $\log(p_{ijt}/p_{ikt}) = v_{ijt} - v_{ikt}$, so the offer-specific representation is equivalent to a binary logit using within-task attribute differences $X_{ijt} - X_{ikt}$. The Stata conditional-logit implementation and an independent maximum-likelihood implementation on these differences agree on the primary coefficients to within $10^{-5}$. The design matrix of within-task differences

has full column rank for each retained specification. That algebraic condition permits estimation under the model; it is not a guarantee of broad experimental support for every counterfactual.

The mixed specification sets $\alpha_i = \bar{\alpha} + \sigma_\alpha z_{\alpha i}$ and $\kappa_i = \bar{\kappa} + \sigma_\kappa z_{\kappa i}$, where the two $z$ terms are independent standard normal variables. The expenditure coefficient remains common. The panel mixed likelihood is

$$L_i(\vartheta) = \int\int \prod_{t=1}^{T_i} \prod_{j \in \mathcal{J}_{it}} p_{ijt}(z; \vartheta)^{y_{ijt}} \phi(z_\alpha)\phi(z_\kappa)\, dz_\alpha dz_\kappa.$$

The same draws affect all of an owner's choices, so integration follows the product over tasks. Integrating task by task would fail to represent persistent preferences. Estimation uses Gauss–Hermite quadrature with 17 and 25 nodes per dimension. The uncertainty calculation uses the inverse numerical Hessian around the optimum and the outer products of respondent likelihood scores.

For a sequential assignment rule depending on recorded history $H_{i,t-1}$, the joint likelihood also contains factors for assignment. These factors can be omitted for preference estimation only when they carry no additional dependence on the unknown preference parameters conditional on that history. Dependence on unrecorded current shocks would violate the maintained interpretation. Eligibility for each common comparison is defined by the displayed receipts; it is not assumed independent of respondent characteristics. The results are consequently conditional preference estimates, with a homogeneous model and a specified random-effects model reported together.

## E.1 Citizen BACE design utility and units

The citizen configuration normalises payments by each respondent's reference bill $Q$. With initial bill $I$, final official bill $T$ and informal payment $B$, set

$$C = T + B, \qquad c = 100C/Q, \qquad e = 100(I - Q)/Q, \qquad o = \max(e, 0)/15.$$

Here $c$ is expenditure in reference-bill percentage points, $e$ is signed initial error on that scale, and $o = 1$ at a 15-point overassessment. With $S$ indicating satellite assessment and $D$ a positive informal payment, engine utility and choice probability are

$$u = -c + \beta S - m_0 D + m_1 D o, \qquad \Pr(\text{option B chosen}) = \text{logit}^{-1}\{\mu(u_B - u_A)\}.$$

The monetary coefficient is fixed at $-1$. Thus $\beta$, $m_0$ and $m_1$ are already in reference-bill percentage-point units, while $\mu$ controls choice consistency. Transaction cost is $m_0 - m_1$ at $e = 15$; $m_1$ is its change from zero to 15-point overassessment, not the cost at that state. Rupee conversion requires the respondent-specific multiplier $Q/100$. Division by $\mu$ is not a monetary conversion.

The executable citizen setup draws Normal priors with mean and standard deviation $(-0.2, 2)$ for $\beta$, $(5.0, 5)$ for $m_0$ and $(-0.2, 5)$ for $m_1$, with initial $\mu$ uniform on $[0.65, 1.65]$. This specification governs adaptive selection and engine updating. The pooled and panel mixed-logit results in Table 6 instead estimate equation (4); their coefficients are in model-specific logit units and expenditure is measured in Rs 1,000. Opening choices remain observed responses, not fitted individual thresholds.

## E.2 Model restrictions and conditional comparisons

For a positive scale $\sigma$ on the conceptual rupee utility, the coefficient mapping is

$$b_p = -1000\sigma, \quad \alpha = \sigma\beta(0), \quad \kappa = -\sigma m(0), \quad \gamma_e = \sigma h(e),$$

$$\delta_e = \sigma[\beta(e) - \beta(0)], \qquad \lambda_e = -\sigma[m(e) - m(0)], \quad e \in \{O, U\}.$$

This mapping explains the signs used in the main text. The ex-post coefficients have model-specific logit units and are distinct from the citizen engine's reference-bill percentage-point parameters.

Table 7 and Figure 2 connect the estimated equation to the restrictions in Section 3. For a vector of restrictions $Rb = 0$, the Wald statistic is

$$W = (Rb)'(R\widehat{\Sigma}R')^{-1}(Rb),$$

with a chi-squared reference distribution whose degrees of freedom equal the number of independent restrictions. A linear comparison $a'b$ has standard error $\sqrt{a'\widehat{\Sigma}a}$; the table reports normal-reference 95% intervals. Here $b$ collects the coefficients and $\widehat{\Sigma}$ is their covariance matrix, using respondent clustering or respondent likelihood scores. The calculations use the saved full-precision estimates and covariance matrices corresponding to the reported models. The reported tests are two-sided and unadjusted for multiple comparisons.

In the mixed model, $\kappa$ denotes the mean paid-deal coefficient. The joint zero-coefficient test therefore concerns mean transaction valuation, not the additional restriction of zero heterogeneity required for savings alone for every owner. State interactions are common across owners in this specification. The contrast $\lambda_O - \lambda_U$ asks whether the change in transaction valuation differs between overassessment and underassessment; a positive $\lambda_O$ alone does not establish that difference.

**Table E.2:** Sensitivity of error interactions and financial responses

| Specification | Satellite × over | Satellite × under | Cash term |
|---|---|---|---|
| Full sequence | 0.093 (0.088) | 0.142* (0.082) | -0.097*** (0.017) |
| Later tasks | -0.065 (0.159) | 0.001 (0.154) | -0.115*** (0.022) |
| Omit extra task | 0.093 (0.088) | 0.145* (0.082) | -0.097*** (0.017) |
| Omit option position | -0.007 (0.087) | 0.045 (0.081) | -0.088*** (0.016) |
| Add requested payment | 0.085 (0.088) | 0.136* (0.082) | 0.028*** (0.007) |
| Rupee error magnitudes | -0.050 (0.039) | -0.024 (0.024) | -0.077*** (0.016) |

*Notes:* Standard errors are respondent clustered. In the error-magnitude row, interactions are per Rs 1,000 of error; other rows use error indicators. The final column is total expenditure per Rs 1,000, except in the additional-payment row, which reports the requested payment conditional on total expenditure. In the separate-component model (not tabulated), the payment coefficient is −0.032 (0.029) and the official-reduction coefficient is 0.065 (0.021), both per Rs 1,000. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided normal-reference Wald tests using the reported standard errors. Stars refer to individual coefficients, not joint tests.

The sensitivity table separates alternative expenditure and error specifications instead of treating all coefficients as the same estimand. Magnitude specifications measure effects per Rs 1,000 of error; state-indicator specifications compare an error state with a correct bill. The separate-component model uses payment and reduction without total expenditure. The additional-payment model conditions on the same total, so changing its payment requires an offsetting change elsewhere in the receipt.

As discussed in Section 6.2, the baseline payoff function has no additional dependence on the size of a positive bribe conditional on expenditure and state. The added requested-payment coefficient is 0.028 (standard error 0.007; two-sided $p < 0.001$), providing evidence against that exclusion within the augmented specification. The response can concern the composition or presentation of expenditure; it does not identify a unique mechanism. Allowing $m_i(e, B)$ would relax the exclusion, while the baseline retains $m_i(e)$ to organise the main conditional comparisons.

# F Additional evidence results

**Table F.1:** Posted prices and purchase decisions

| Term | Percentage points (SE) | 95% interval |
|---|---|---|
| Free offer | 94.76*** (0.86) | [93.07, 96.45] |
| Rs 100 versus free | -55.75*** (2.01) | [-59.68, -51.82] |
| Rs 200 versus free | -55.36*** (2.04) | [-59.35, -51.36] |

*Notes:* Saturated linear probability model, 2,099 observed responses. HC3 standard errors and two-sided t-reference inference. * p<0.10, ** p<0.05, *** p<0.01. The free-offer constant is the purchase share; other coefficients compare each paid offer with free provision.

The posted-price comparison retains 668 valid responses at zero, 728 at Rs 100 and 703 at Rs 200. One additional owner in the Rs 200 group supplies no substantive purchase answer. The saturated model reports each paid-price contrast separately. It does not require willingness to pay to follow a specified distribution, and the lack of a further decline between the two paid offers is visible in the reported shares.

**Table F.2:** Incremental information contrasts

| Outcome | *N* | T1–control | T2–control | T2–T1 |
|---|---|---|---|---|
| Correction easy / very easy | 2,032 | 1.68<br>[-3.02, 6.39] | 0.84<br>[-3.83, 5.50] | -0.85<br>[-4.28, 2.58] |
| Expected accurate houses (0–10) | 2,073 | 0.09<br>[-0.15, 0.32] | 0.16<br>[-0.08, 0.39] | 0.07<br>[-0.10, 0.25] |
| Accept independent information | 2,099 | 2.07<br>[-4.35, 8.50] | 0.05<br>[-6.37, 6.47] | -2.03<br>[-6.62, 2.57] |
| Opening satellite choice | 2,100 | 2.66<br>[-3.73, 9.04] | 1.76<br>[-4.61, 8.14] | -0.90<br>[-5.44, 3.65] |
| Paid inspector acceptance | 2,050 | 1.92<br>[-4.61, 8.45] | -1.31<br>[-7.84, 5.22] | -3.23<br>[-7.90, 1.44] |

Entries are unadjusted differences and pointwise 95% HC1 intervals, in percentage points except accuracy (number out of ten). Each item uses its substantive-answer sample. All groups receive record-update and appeal information; T1 adds satellite information and T2 also shows an estimate for a comparable property. Assignment is not independently verified in the saved analysis. These exploratory contrasts are reported as group comparisons. Easy/very easy is coded one; accuracy retains its count scale. The separate 399-interview wave is not included.